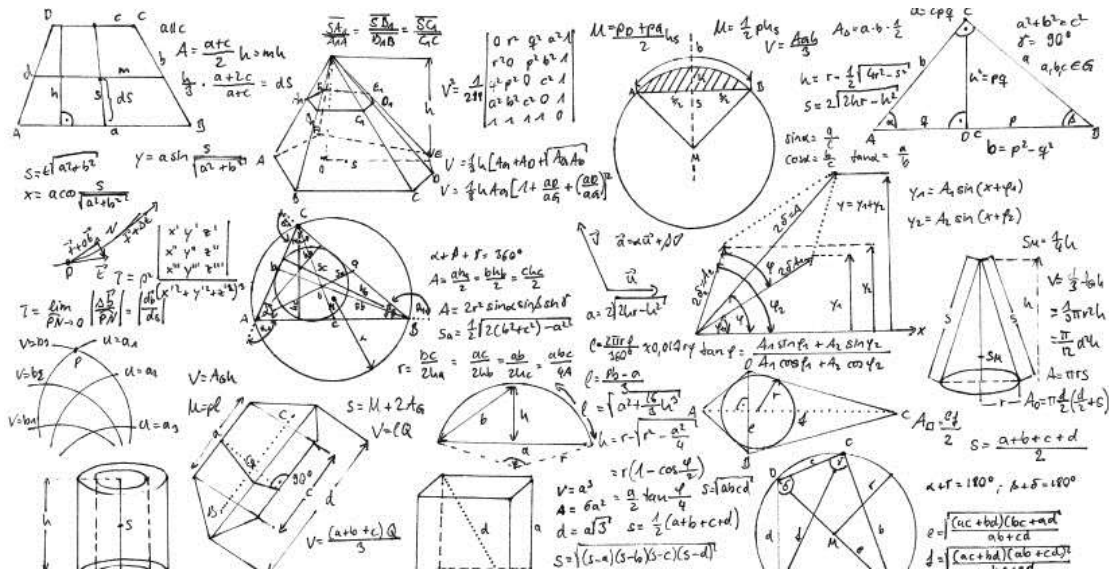


IKAROS



AVIATION TRAINING CENTER CY.147.001



EASA PART 66 MODULE 01 COMMON MATHEMATICS

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MODULE 02 - PHYSICS

Part 66 Ref

1.1 Arithmetic

Arithmetical terms and signs, methods of multiplication and division, fractions and decimals, factors and multiples, weights, measures and conversion factors, ratio and proportion, averages and percentages, areas and volumes, squares, cubes, square and cube roots.

1.2 Algebra

a) Evaluating simple algebraic expressions, addition, subtraction, multiplication and division, use of brackets, simple algebraic fractions; Linear equations and their solutions; Simultaneous equations and second degree equations with one unknown.

1.3 Numbers

Indices and powers, negative and fractional indices; Binary and other applicable numbering systems; Logarithms;

1.4 Geometry

Simple geometrical constructions; triangle, polygon, quadrilaterals, parallelogram, rectangle, rhombus, square, trapezium, circles; Area and volume.

1.5 Graphs

Graphical representation; nature and uses of graphs, graphs of equations/functions.

1.6 Trigonometry

Simple trigonometry; trigonometrical relationships, use of tables and rectangular and polar coordinates.

Extra Reading References

AMT Mechanical Technicians Handbook

Dale Clark - Dictionary of Aircraft Terms

EASA Module 01 Mathematics – Amazon.com

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$$\begin{array}{r} 147 \\ + 65 \\ \hline \end{array}$$

$$\begin{array}{r} 249 \\ + 54 \\ \hline \end{array}$$

$$\begin{array}{r} 480 \\ + 96 \\ \hline \end{array}$$

$$\begin{array}{r} 724 \\ + 77 \\ \hline \end{array}$$

$$\begin{array}{r} 338 \\ + 164 \\ \hline \end{array}$$

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$$\begin{array}{r} 328 \\ - 68 \\ \hline \end{array}$$

$$\begin{array}{r} 169 \\ - 74 \\ \hline \end{array}$$

$$\begin{array}{r} 943 \\ - 52 \\ \hline \end{array}$$

$$\begin{array}{r} 517 \\ - 63 \\ \hline \end{array}$$

$$\begin{array}{r} 836 \\ - 268 \\ \hline \end{array}$$

$$\begin{array}{r} 545 \\ - 385 \\ \hline \end{array}$$

$$\begin{array}{r} 905 \\ - 628 \\ \hline \end{array}$$

$$\begin{array}{r} 734 \\ - 461 \\ \hline \end{array}$$

PART 66 MODULE 6 B1/B2

SECTION 01

ARITHMETIC

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ARITHMETIC

Introduction

Mathematics is the basic language of science and technology. It is an exact language that has a vocabulary and meaning for every term. Since mathematics follows definite rules and behaves in the same way every time, scientists and engineers use it as their basic tool.

Long before any metal is cut for a new aircraft design, there are literally millions of mathematical computations made. Aviation maintenance technicians perform their duties with the aid of many different tools. Like the wrench or screwdriver, mathematics is an essential tool in the maintenance, repair and fabrication of replacement parts.

With this in mind, you can see why you must be competent in mathematics to an acceptable level. These notes cover the complete mathematics syllabus required to comply with the JAR-66 B1 and B2 licence level.

Arithmetic is the basic language of all mathematics and uses real, non-negative numbers. These are sometimes known as counting numbers. Only four operations are used, **addition, subtraction, multiplication** and **division**.

Whilst these operations are well known to you, a review of the terms and operations used will make learning the more difficult mathematical concepts easier.

Arithmetic Terms

The most common system of numbers in use is the decimal system, which uses the ten digits 0, 1, 2, 3, 4, 5, 6, 7, 8, 9.

These ten whole numbers from zero to 9 are called '**Integers**'. Above the number nine, the digits are re-used in various combinations to represent larger numbers. This is accomplished by arranging the numbers in columns based on a multiple of ten. With the addition of a **minus (-)** sign, numbers smaller than zero are indicated.

To describe quantities that fall between whole numbers, fractions are used. Common fractions are used when the space between two integers is divided into equal segments, such as quarters. When the space between integers is divided into ten segments, decimal fractions are typically used.

Students will be familiar with this system and the **basic operations**, which may involve **Addition, Subtraction, Multiplication and Division**.

- **When numbers are added, they form a sum.**
- **When numbers are subtracted, they create a difference.**
- **When numbers are multiplied, they form a product.**
- **When one number (the dividend) is divided by another (the divisor), the result is a quotient.**

It is useful if a student is proficient at simple mental arithmetic, and this is only possible if one has a "feel" for numbers, and the size of numbers. A knowledge of **simple "times tables"** is also useful.

0	1	2	3	4	5	6	7	8	9	10
1	1	2	3	4	5	6	7	8	9	10
2	2	4	6	8	10	12	14	16	18	20
3	3	6	9	12	15	18	21	24	27	30
4	4	8	12	16	20	24	28	32	36	40
5	5	10	15	20	25	30	35	40	45	50
6	6	12	18	24	30	36	42	48	54	60
7	7	14	21	28	35	42	49	56	63	70
8	8	16	24	32	40	48	56	64	72	80
9	9	18	27	36	45	54	63	72	81	90
10	10	20	30	40	50	60	70	80	90	100

Fig 01. 1 - Times Table

The following simple tests for divisibility may be useful. A number is divisible by:

- **2** if it is an even number.
- **3** if the sum of the digits that form the number is divisible by 3.
- **4** if the last two digits are divisible by 4.
- **5** If the last digit is 0 or 5.
- **10** if the last digit is 0

Directed or Signed Numbers

Directed numbers are numbers which have a **+** or **- sign** attached to them.

Directed numbers can be added, subtracted, etc., but care should be taken to ensure a correct solution. The following rules should assist.

- To add several numbers of the same sign, add them together and ensure sign of the sum is the same as the sign of the numbers.
- To add 2 numbers with different signs, subtract the smaller from the larger. The sign of the resultant (the difference) is the same as the sign of the large number.

eg. For example $-12 + 6$, $(12 - 6) = 6$, $-12 + 6 = -6$

If there are more than 2 numbers, carry out the operation 2 numbers at a time, or produce two numbers by adding up all the numbers with like signs. And then apply the rules above.

Example:

$$-15 - 8 + 13 - 19 + 6, (-15 - 8) = -23, -23 + 13 = -10, -10 - 19 = -29, -29 + 6 = -23$$

$$\text{or } -15 + (-8) + (-19) = -42 \text{ and } +13 + 6 = +19$$

$$-42 + 19 = -23$$

To subtract directed numbers, change the sign of the number to be subtracted and add the resulting numbers.

Example:

$$\begin{array}{rclclcl} -10 - (-6) & = & -10 + 6 & = & -4 \\ 7 - (+18) & = & 7 - 18 & = & -11 \end{array}$$

A minus in front of brackets should be taken to mean **-1**. Using the above example **-(6)** should be read as **-1(-6)** i.e. **minus 1 times minus six**. Similarly, a positive sign in front of brackets should be read as **+1**, so **+(-6)** should be read as **+1(-6)** i.e. **plus 1 times minus 6**.

- **The product of two numbers with like signs is positive (+ve)**
- **The product of numbers with unlike signs is negative (-ve).**
- **When dividing numbers with like signs, the quotient of the result is +ve.**
- **When dividing numbers with unlike signs, the quotient is -ve.**

This can be summarised as follows:

$$\begin{aligned} (+) \times (+) &= (+) \\ (+) \times (-) &= (-) \end{aligned}$$

$$\begin{aligned} (-) \times (+) &= (-) \\ (-) \times (-) &= (+) \end{aligned}$$

FACTORS

We know that $2 \times 6 = 12$. 2 and 6 are factors of 12.

We could also state that, as $3 \times 4 = 12$, 3 and 4 are also factors of 12, similarly 12 and 1.

This may seem obvious, but it is sometimes useful to "**factorise**" a number, i.e. determine the factors that make up the number. More commonly it is necessary to find the factors of an algebraic expression.

Example

Find the possible factors of 60.

(in other words, find the integers that divide 60). The factors will be:

2, 3, 4, 5, 6, 10, 12, 15, 20, 30 and 60

Check them yourself.

PRIME NUMBERS

A prime number is a number whose only factors are 1 and itself. The prime numbers between 1 and 30 are:

1, 2, 3, 5, 7, 11, 13, 17, 19, 23 and 29. Check them yourself.

It is sometimes useful to express the factors of a given number in terms of prime numbers.

For example, let us look at the factors of 60 again, taking 4 and 15 as 2 factors.

($4 \times 15 = 60$), but 4 has factors of 2 and 2, and 15 has factors of 5 and 3.

Hence the number **60** can be expressed as $2 \times 2 \times 3 \times 5$, which are all factors of **60**.

Note

We have now factorised the number 60 in terms of prime numbers.

GREATEST COMMON FACTOR (GCF)

The greatest common factor is the biggest factor (number) that divides the numbers being examined. Suppose that we take 3 numbers, **1764**, **2100** and **2940**. The greatest common factor of these numbers is 84.

In some instances you will be able to identify this value simply by looking at the numbers, in others you will need to calculate it. To calculate the **GCF**, we must identify the factors of each number in terms of prime numbers:

$$\begin{array}{l}
 \boxed{2 \times 2 \times 3 \times 3 \times 7 \times 7} = 1764 \\
 \boxed{2 \times 2 \times 3 \times 5 \times 5 \times 7} = 2100 \\
 \boxed{2 \times 2 \times 3 \times 5 \times 7 \times 7} = 2940
 \end{array}$$

We then select the common prime factors and multiply them together to produce the Greatest Common Factor, in this case:

$$2 \times 2 \times 3 \times 7 = 84$$

LOWEST COMMON MULTIPLE (LCM)

The lowest common multiple of a set of numbers is the smallest number that is divided exactly by each of the given numbers. The **LCM** can be found by multiplying together all of the factors the maximum number of that appear to each of the individual numbers.

Consider the previous three numbers, **1764**, **2100** and **2940** and their factors.

$$\begin{array}{l}
 \boxed{2 \times 2} \times \boxed{3 \times 3} \times \boxed{7 \times 7} = 1764 \\
 \boxed{2 \times 2} \times 3 \times \boxed{5 \times 5} \times 7 = 2100 \\
 \boxed{2 \times 2} \times 3 \times 5 \times 7 \times 7 = 2940
 \end{array}$$

The Lowest Common Multiple of these three numbers will be:

$$2 \times 2 \text{ (in all)} \times 3 \times 3 \text{ (in 1764)} \times 5 \times 5 \text{ (in 2100)} \times 7 \times 7 \text{ (in 1764 and 2940)}$$

$$2 \times 2 \times 3 \times 3 \times 5 \times 5 \times 7 \times 7 = 44,100$$

$$\text{so } 2 \times 2 \times 3 \times 3 \times 5 \times 5 \times 7 \times 7 = 44,100 \text{ is the LCM}$$

$$1764 \times 25 = 44,100$$

$$2100 \times 21 = 44,100$$

$$2940 \times 15 = 44,100$$

ARITHMETICAL PRECEDENCE

The term Arithmetic Precedence means the order in which we carry out arithmetic functions. Sometimes it doesn't matter what order we carry them out.

Consider the expression $2 + 3 = 5$. It makes no difference if we write $3 + 2 = 5$.

Again, consider $3 \times 4 = 12$, there is no difference if we write $4 \times 3 = 12$.

However, if I write $2 + 3 \times 4$, what is the answer?

- If we first add $2 + 3$, we will get 5 and then $5 \times 4 = 20$.
- Alternatively, multiplying $3 \times 4 = 12$, adding 2 we get 14.

If we are going to agree on the answer we must first agree on the rules we use.

This introduces the topic known as arithmetical precedence, and is most easily remember by the term **BODMAS**.

BODMAS indicates the precedence, or the order in which we perform our calculations:

- **B** stands for Brackets
- **O** stands for "Of"
- **D** stands for Division
- **M** stands for Multiplication
- **A** stands for Addition
- **S** stands for Subtraction

BODMAS Example

Find the value of: $64 \div (-16) + (-7 - 12) - (-29 + 36) (-2 + 9)$

This expression becomes:

$$\begin{array}{rcl}
 64 \div (-16) + (-19) - (7)(7) & & \text{B} \\
 (-4) + (-19) - (7)(7) & & \text{D} \\
 (-4) + (-19) - 49 & & \text{M} \\
 -23 - 49 & & \text{A} \\
 = -72 & & \text{S}
 \end{array}$$

FRACTIONS

$$\frac{11}{16}$$

is an example of a Proper Fraction, generally abbreviated to fraction.

It has the same meaning as $11 \div 16$, that is, 11 divided by 16.

- The number above the line is the Numerator
- The number below the line is the Denominator.

$$\frac{23}{4}$$

Is also a Fraction, but because **23** is greater than **4**, it is called an **Improper Fraction**. It will normally be written as $5 \frac{3}{4}$ which is the same as:

$$\frac{23}{4} = \frac{20 + 3}{4} = \frac{20}{4} + \frac{3}{4} = 5 \frac{3}{4}$$

Similarly;

$$3 \frac{4}{7} \text{ could be converted to } \frac{25}{7} \text{ because } 3 \times 7 = \frac{21}{7}$$

$$\text{so } \frac{21}{7} + \frac{4}{7} = \frac{25}{7}$$

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Addition of Fractions

The important thing to remember here is that only fractions with the same (a common) denominator can be added or subtracted.

Example 1

$$\frac{7}{8} + \frac{3}{8} + \frac{1}{8} = \frac{11}{8} = 1\frac{3}{8}$$

If the denominators are not the same, then it is necessary to find the **lowest Common Denominator (LCD)** and to put each fraction in terms of this value. Finding the **Lowest Common Denominator** is essentially the same as finding the **Lowest Common Multiple**, which was covered in a previous topic.

Example 2:

$$\frac{7}{16} + \frac{5}{12} + \frac{1}{8}$$

In this example, the **LCD of 16, 12 and 8 is 48**. In some cases it may be quicker to find a common denominator by simply multiplying the denominators together i.e. **16 x 12 x 8 = 1536**.

Note, this is not the LCD.

Having found the **LCD**, each fraction now needs to be expressed in terms of the **LCD**. This is achieved by dividing the **LCD** by the denominator and multiplying the result by the numerator.

$\frac{7}{16}$ Divide the LCD by the denominator $48 \div 16 = 3$
 Multiply the result by the numerator $3 \times 7 = 21$
 and so $\frac{7}{16}$ can be written as $\frac{21}{48}$

Alternatively:

Divide the LCD by the denominator $48 \div 16 = 3$
 And multiply top and bottom of the fraction by the result

$\frac{7}{16} \times \frac{3}{3} = \frac{21}{48}$ so $\frac{21}{48}$ is the same as $\frac{7}{16}$

Similarly:

$\frac{5}{12}$ Divide the LCD by the denominator $48 \div 12 = 4$
 Multiply top and bottom of fraction by 4 = $\frac{5}{12} \times \frac{4}{4} = \frac{20}{48}$

Finally

$\frac{1}{8}$ converts to $\frac{1}{8} \times \frac{6}{6} = \frac{6}{48}$

so the three fractions $\frac{7}{16} + \frac{5}{12} + \frac{1}{8}$ become $\frac{21}{48} + \frac{20}{48} + \frac{6}{48}$

with the common denominator in place, the addition becomes $\frac{47}{48}$

Example 3

In this case we have 3 improper fractions

$$1\frac{2}{3} + 2\frac{5}{6} + 3\frac{7}{12}$$

First add the whole numbers together, so the calculation becomes.

$$6 + \frac{2}{3} + \frac{5}{6} + \frac{7}{12}$$

The LCD of 3, 6 and 12 is 12. Using this, the sum becomes.

$$\begin{aligned} & 6 + \frac{2(4)}{3(4)} + \frac{5(2)}{6(2)} + \frac{7(1)}{12(1)} \\ &= 6 + \frac{8}{12} + \frac{10}{12} + \frac{7}{12} \\ &= 6 + \frac{25}{12} \end{aligned}$$

This simplifies to become:

$$6 + 2 + \frac{1}{2} = 8\frac{1}{2}$$

Subtraction of Fractions

The basic procedure is very similar to that used for addition; find the LCM, convert the individual fractions, but subtract the numerators instead of adding. There may be one difference which is important.

Example 1

$$3\frac{1}{3} - 1\frac{7}{12}$$

Subtract the whole numbers, $3 - 1 = 2$, so the calculation becomes:

$$2 + \frac{1}{3} - \frac{7}{12}$$

The LCM is 12, so the sum becomes:

$$2 + \frac{4}{12} - \frac{7}{12}$$

Now $\frac{7}{12}$ is greater than $\frac{4}{12}$ and so $\frac{12}{12} (= 1)$ is “borrowed” from the 2 so

$$2 + \frac{1}{3} - \frac{7}{12} \text{ becomes } 1 + \frac{12}{12} + \frac{4}{12} - \frac{7}{12} = 1 + \frac{9}{12} \text{ written as } 1\frac{9}{12}$$

To avoid confusion, you may find it easier to convert the mixed numbers $3\frac{1}{3}$ to improper fractions $\frac{10}{3}$, find the **LCM**, perform the subtraction and then simplify the answer.

Multiplication of Fractions

These calculations are generally easier to perform than addition and subtraction.

Example 1

$$\frac{5}{6} \times \frac{7}{8}$$

Simply multiply the numerators together and multiply the denominators together.

So:

$$\frac{5}{6} \times \frac{7}{8} = \frac{35}{48}$$

And then convert to a mixed number or simplify as necessary.

Example 2

$$1\frac{2}{3} \times 4\frac{1}{4}$$

Convert into improper fractions, so:

$$1\frac{2}{3} \text{ becomes } \frac{5}{3} \text{ and } 4\frac{1}{4} \text{ becomes } \frac{17}{4}$$

Then multiply as before:

$$\frac{5}{3} \times \frac{17}{4} = \frac{85}{12}$$

And convert to a mixed number:

$$7\frac{1}{12}$$

Division of Fractions

To divide two fractions we invert the divisor (the number we are dividing by) and multiply.

Example 1

$$6\frac{1}{8} \div \frac{7}{16}$$

Firstly convert into improper fractions, then invert the second fraction and multiply:

$$\frac{49}{8} \div \frac{7}{16} = \frac{49}{8} \times \frac{16}{7} = \frac{784}{56} = 14$$

NOTE

Every opportunity should be taken to simplify by 'Cancelling' numbers above and below the line wherever possible.

For Example:

$$\frac{49}{8} \div \frac{7}{16} = \frac{7 \times \cancel{7}}{\cancel{8}} \times \frac{\cancel{8} \times 2}{\cancel{7}}$$

Which becomes:

$$7 \times 2 = 14$$

(a 7 above and below the line cancels, as does an 8).

DECIMAL FRACTIONS

Decimal fractions are fractions where the Denominator is equal to some power of 10, i.e. 100, 1,000, 10,000 etc.

The example below is a decimal fraction:

$$\frac{125}{1000}$$

Decimal fractions are usually re-written as decimals. This is very easily done by using a Decimal Point. Take the example above:

Place a decimal point to the right of the numerator (top number). Then move the decimal point to the left, by a number of places equal to the number of "noughts" in the denominator (bottom number). Remove one nought from the denominator for each move.

So $\frac{125}{1000}$ starts as $\frac{125.}{1000}$ becomes $\frac{12.5}{100}$ then $\frac{1.25}{10}$ and finally $\frac{0.125}{1}$

Any value over **1** is equal to that value so the answer becomes 0.125

$\frac{125}{10}$ would become 12.5 etc.

Any fraction can be formed into a decimal, by dividing the numerator by the denominator. For example:

$\frac{7}{8}$ becomes 0.875

This is found through a process of long division.

Addition & Subtraction

The main thing to remember when adding or subtracting decimal numbers is to ensure they are correctly lined up using the decimal point as a reference.

Example 1

$$2.683 + 34.41$$

$$\begin{array}{r} 2.683 \\ 34.410 \\ 37.093 \end{array}$$

the answer is 37.093

Multiplication & Division

Multiplication of Decimals is the same as ordinary "long" multiplication, but the number of decimal places in the answer must equal the sum of decimal places in the numbers being multiplied.

Example 1

$$6.24 \times 3.121$$

$$\begin{array}{r}
 24 \\
 3.121 \\
 \hline
 624 \\
 12480 \\
 62400 \\
 1872000 \\
 \hline
 1947504
 \end{array}$$

There are two digits after the decimal place in the first number and 3 in the second. Therefore, there must be **5 digits** after the decimal place in the answer, so the answer becomes:

Answer is 19.47504

(Common sense helps here. A number slightly greater than 6 is multiplied by another number slightly greater than 3. Logically the answer should be approximately 18).

Division is also the same as ordinary long division, but again a simple rule helps to simplify the process

'Do not try to divide by a fraction'

Multiply both the divisor and dividend by a power of ten (move the decimal place to the right) so that the divisor becomes a whole number.

Examples:

$$3650 \div 45.56$$

Multiply both numbers by 100 (10^2)

to give $365000 \div 4556$

$$769 \div 0.364$$

Multiply both numbers by 1000 (10^3)

to give $769000 \div 364$

WEIGHTS AND MEASURES

A wide number of different weights and measures are used during the maintenance of aircraft. The ones that come to mind first are probably fuel capacities, tyre pressures, temperatures and speeds. There are however very many others, which you will meet as you progress through your course.

Firstly, the most commonly used system in aviation today is the '**Systeme Internationale**' (**SI**). This system is based on multiples of **10** and has been accepted widely, with one or two exceptions.

It consists of a standard set of units for:

- **Length (metre)**
- **Mass (kilogram)**
- **Time (second)**
- **Temperature (Kelvin)**
- **Current (ampere)**
- **Light (candela)**

There are several other units which, whilst not being part of the basic **S.I.** ones above, are in common use and still use the metric system for calculations.

An older system that is still used in some countries today, is the **Imperial System**, which uses a mixture of old units such as

- **Length (feet and inches)**
- **Weight (pounds)**
- **Capacity (gallons)**
- **Temperature (Fahrenheit)**

You will occasionally meet a mixture of systems, which will require conversion from one to another. A good example is the amount of fuel put into an aircraft's tanks. You will find this being measured in imperial gallons, American gallons, imperial pounds, SI kilograms or metric litres.

Changing a quantity in one unit to a quantity in another unit requires a conversion factor. When the quantity in the first unit is multiplied by the conversion factor, the result is the quantity in the second units. For example, to convert imperial gallons to litres, they must be multiplied by **4.546**

Example 1

Convert 25 Imperial Gallons into Litres.

$$25 \times 4.546 = 113.65 \text{ Litres.}$$

Example 2

Convert 1500 miles into kilometres using the conversion factor **1.6094**

$$1500 \times 1.6094 = 2413.9 \text{ Kilometres}$$

Note

You will normally be given the conversion factor, however, you may have to transpose a formula in order to use it.

RATIO AND PROPORTION

This topic is an extension of several previous topics. **'Ratio' and 'Proportion'** are essentially statements that link two or more **"Quantities"** together. For example, a **'3 to 1 mix of sand and cement'**, which may be written as a 3:1 mix of sand and cement, means 'mix 3 parts of sand to 1 part of cement'.

This is a commonly used statement which you will notice has no formal units, although volume is inferred. Parts could be represented by shovels full, buckets full, wheelbarrows full etc.

The mixture simply has a total of 4 parts, of which 3 parts, or **$\frac{3}{4}$** , is sand, and 1 part $\frac{1}{4}$ is cement. A ratio therefore simply provides a means of comparing one value with another.

For example, if an engine turns at **4000rpm** and the propeller turns at **2400rpm**, the ratio of the two speeds is **4000 to 2400**, or **'5 to 3'** when reduced to its lowest terms. This relationship can also be expressed as **$\frac{5}{3}$ or 5:3**.

The use of **'Ratios'** is common in aviation, such as when considering the compression ratio in an engine. This is the ratio of cylinder displacement, when the piston is at the bottom of its stroke compared with the displacement when it is at the top.

For example, if the volume of the cylinder at the bottom of its stroke is **240 cm³** and at the top becomes **30 cm³** the ratio is **240:30** or, reduced to its lowest terms, **8:1**.

Another typical ratio is that of different gear sizes. For example, the ratio of a drive gear with **15 teeth** to a driven gear with **45 teeth** is **15:45 or 1:3** when reduced. This means that for every one tooth of the drive gear there are three teeth on the driven gear.

However, when working with gears, the ratio of teeth is opposite the ratio of revolutions. In other words, since the drive gear has one third as many teeth as the driven gear, the drive gear must complete three revolutions to turn the driven gear once. This results in a revolution ratio of **3:1**, which is the opposite of the ratio of teeth.

A '**Proportion**' is a statement of equality between two or more ratios and represents a convenient way to solve problems involving ratios. For example, if an engine has a reduction gear ratio between the crankshaft and the propeller of **3:2**, and the engine is turning at **2700rpm**, what is the rotational speed of the propeller?

In this problem let V_p represent the unknown value, which in this case is the speed of the propeller. Next, set up a proportional statement using the fractional form, $\frac{3}{2} = \frac{2700}{V_p}$.

To solve this equation, cross multiply to arrive at the equation $3V_p = 2 \times 2700$, or **5400rpm**. To solve for V_p divide **5400** by **3**. Thus, the propeller speed is **1800rpm**.

Example

Divide **£240** between **4 men** in the ratio of **9:11:13:15**.

The normal procedure for this type of problem is to:

- **Add all of the individual proportions to find the total number of parts.**
- **Divide the total amount by the number of parts to find the value of each part.**
- **Multiply each ratio by the value of each part.**

$$\text{So. } 9 + 11 + 13 + 15 = 48$$

$$\text{£240 divided by } 48 = \text{£5}$$

Therefore each part is worth £5

$$9 \times 5 = 45$$

$$11 \times 5 = 55$$

$$13 \times 5 = 65$$

$$15 \times 5 = 75$$

The proportions are therefore **£45, £55, £65** and **£75**

A useful check is to add the individual parts together, to ensure the total is the amount you started with.

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AVERAGES AND PERCENTAGES

Averages

When working with numerical information, it is sometimes useful to find the average value. When estimating the time of a particular journey there would be no point in basing the time on the slowest speed or the highest speed, always use an average speed.

We would also use average fuel consumption to estimate how much fuel an aircraft would use for a particular flight.

In both of these types of calculation, we can only work out the average by dividing the total distance or fuel used by the time.

Example 1

An aircraft travels a total distance of **750 km** in a time of **3 hours 45 minutes**. What is the average speed in **km/hr**?

$$\begin{aligned}\text{Average Speed} &= \text{Total Distance/Time} \\ &= \frac{750}{3.75} = 200 \text{ km/hr}\end{aligned}$$

Example 2

An aircraft uses **300 gallons** of fuel for a flight of duration **4 hours**. What is the average fuel consumption?

$$\text{Average Fuel Consumption} = \frac{300}{4} = 75 \text{ gallons/hour}$$

We often need to calculate averages based on many data items.

Example 3

The weight of six items are as follows:

9.5, 10.3, 8.9, 9.4, 11.2, 10.1

What is the average weight?

To calculate this we simply add the total weights and divide by the number of items.

$$\begin{array}{r} 9.5 \\ 10.3 \\ 8.9 \\ 9.4 \\ 11.2 \\ \underline{10.1} \\ \text{Total} = 59.4\text{kg} \div 6 \\ \text{Average} = 9.9\text{kg} \end{array}$$

PERCENTAGE

Introduction

Percentages are special fractions whose denominator is **100**. The decimal fraction **0.33** is the same as **33/100** and is equivalent to **33 percent** or **33%**. You can convert common fractions to percentages by first converting them to decimal fractions and then multiplying by **100**.

For example:

5/8 expressed as a decimal is 0.625

This is converted to a percentage by moving the decimal point two places to the right, the same as multiplying by **100**. This becomes **62.5%**.

To find the percentage of a number, multiply the number by the decimal equivalent of the percentage.

Example1

To find **10%** of **200**, begin by converting **10%** to its decimal equivalent, which is **0.1**. This is achieved by dividing the percentage figure by **100**. Now multiply **200** by **0.1** to arrive at the value of **20**.

If you want to find the percentage one number is of another, you must divide the first number by the second and multiply the quotient by **100**. For instance:

An engine produces **85hp** from a possible **125hp**. What percentage of the total horsepower available is being developed? To solve this, divide the **85** by **125** and multiply the quotient by **100**.

$$\begin{aligned} 85 \div 125 &= 0.68 \\ 0.68 \times 100 &= 68\% \text{ power} \end{aligned}$$

Example 2

Another way that percentages are used is to determine a number when only a portion of the number is known. For example:

If **4180rpm** is **38%** of the maximum speed, what is the maximum speed?

To determine this, you must divide the known quantity, **4180rpm**, by the decimal equivalent of the percentage.

$$4180 \div 0.38 = 11,000 \text{ rpm maximum}$$

A common mistake made on this type of problem is multiplying by the percentage instead of dividing. One way of avoiding making this error is to look at the problem and determine what exactly is being asked.

In the problem above, if **4180rpm** is **38%** of the maximum then the maximum must be greater than **4180**. The only way to get an answer that meets this criterion is to divide by **0.38**.

POWERS AND ROOTS

Introduction

A root is a number that when multiplied by itself a specified number of times will produce a given number.

The two most common roots are the '**Square Root**' and the '**Cube Root**'.

Square Roots

The **square root of 25**, written as $\sqrt{25}$, equals 5. That is, when the number **5** is squared (multiplied by itself), it produces the number **25**. The symbol $\sqrt{}$ is called a radical sign. Finding the square root of a number is the most common application of roots.

Numbers whose square roots are whole numbers are called perfect squares. The first ten perfect squares are: 1, 4, 9, 16, 25, 36, 49, 64, 81, and 100. The square root of each of these numbers is 1, 2, 3, 4, 5, 6, 7, 8, 9, and 10, respectively.

Examples:

$$\sqrt{36} = 6 \text{ and } \sqrt{81} = 9$$

To find the square root of a number that is not a perfect square, use either a calculator or the estimation method. A long hand method does exist for finding square roots, but with the advent of calculators and because of its lengthy explanation, it is not included in this module. The estimation method uses the knowledge of perfect squares to approximate the square root of a number.

Example:

Find the **square root of 31**. Since **31** falls between the two perfect roots **25** and **36**, we know that **31** must be between **25** and **36**. Therefore, **31** must be greater than **5** and less than **6** because **25 = 5** and **36 = 6**.

If you estimate the square root of **31** at **5.5**, you are close to the correct answer. The square root of **31** is actually **5.568**.

Cube Roots

The **cube root of 125** equals **5**. written as

$$\sqrt[3]{125}$$

That is, when the number **5** is cubed (**5 × 5 × 5**), it produces the number **125**.

It is common to confuse the "**cube**" of a number with the "**cube root**" of a number. For clarification,

$$\text{The Cube of } 27 = 27^3 = 27 \times 27 \times 27 = 19,683$$

$$\text{The Cube Root of } 27 = \sqrt[3]{27} = 3$$

FRACTIONAL INDICIES

Another way to write a root is to use a fraction as the index (or exponent) instead of the radical sign. The square root of a number is written with a **$\frac{1}{2}$** as the exponent instead of a radical sign.

The cube root of a number is written with an exponent of **$\frac{1}{3}$** and the fourth root with an exponent of **$\frac{1}{4}$** and so on.

Examples:

$$\sqrt{31} = 31^{\frac{1}{2}} \quad \sqrt[3]{125} = 125^{\frac{1}{3}} \quad \sqrt[4]{16} = 16^{\frac{1}{4}}$$

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ALGEBRA

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ALGEBRA

Introduction

Very often students will claim that they never have and never will understand Algebra. They say they can understand and work with numbers, but not with letters, and yet Algebra is designed to make matters simple and clear.

For example, suppose a room is **5 metres** long by **3 metres** wide and we need to know how much carpet is needed to cover the floor. No one would have any hesitation in calculating the answer, **15 square metres (m²)**. But that answer only applies to that room. The general answer is that the area is found by multiplying length by width (or breadth).

i.e. Area = length x breadth

But it is easier to write **A = L x b**, where the letters **A, L, b** represent in this case **Area, Length and breadth**, and that is what Algebra is all about; letters represent some variable and only when particular values. i.e. numbers are known, do we resort to them instead.

So when using Algebra, it is important to state what the letters represent. Some letters are often used, particularly **x and y**, but **g** often represents acceleration due to gravity, **ρ** represents density, and so on. This is what Algebraic notation is about.

Operation

Algebraic operations are in essence the same as when using numbers. So

- Adding a and b is written $a + b$
- Subtracting a and b is written $a - b$
- Multiplying a and b is written ab
- Dividing a by b is written a/b
- Squaring a a^2

We are not restricted to 2 letters only.

a Multiplied by **b** and divided by **c** becomes, logically:

$$\frac{ab}{c}$$

Note also that the order in which letters appear is basically unimportant.

$$a \times b \times c \times d = abcd = bdac = cadb \text{ etc.}$$

(3 x 4 is obviously the same as 4 x 3 etc.)

When symbols such as **x** and **y** are multiplied together we do not need to include the multiplication sign. This is the same if a number and a symbol are multiplied together.

$$3 \times y, 4 \times z, s \times p, a \times b, y \times z \times m$$

can all be written without the multiplication sign as

$$3y, 4z, sp, ab \text{ and } yzm$$

The same is not true of numbers on their own:

$$7 \times 8, 4 \times 5 \text{ and } 6 \times 7 \text{ cannot be written as } 78, 45 \text{ and } 67.$$

‘**Like Terms**’ are terms comprised of the same algebraic quantity - this is important. **7x**, **5x** and **-3x** are all terms containing **x**

7a, **4b**, **3a** and **-6b** can be split into two groups of like terms, **7a** and **3a**, and **4b** and **-6b**.

If like terms contain numerical coefficients, they can be simplified.

$$7x + 5x - 3x = (7 + 5 - 3)x = 9x$$

$$7a + 3a + 4b - 6b = 10a - 2b.$$

Terms like **ab + cb - db** may be simplified as **(a + c - d) b**. (**b** is a common factor of the **3** terms)

When dealing with algebraic terms and expressions the ability to factorise is a great asset. Similarly, the ability to divide numerator and denominator by the same terms (i.e. cancelling top and bottom) allows simplification.

$$\frac{3a^2b}{6ab^2} = \frac{\cancel{3}a\cancel{a}\cancel{b}}{\cancel{6}\cancel{a}\cancel{b}b} = \frac{a}{2b}$$

Basic Laws

Algebra obeys the same laws of procedure as Arithmetic, i.e. **BODMAS**.

Note that Brackets appear rather more often in Algebra, and are only removed when there is a good reason to do so, for example, when further operations ultimately lead to greater simplification.

$$(3x + 7y) - (4x + 3y) = 3x + 7y - 4x - 3y = -x + 4y$$

Note especially that when removing brackets, all the terms inside the brackets are multiplied by what is immediately outside the brackets. The basic procedure is as follows.

$$a(x + y) = ax + ay$$

$$a + b(x + y) = a + bx + by \text{ (both } x \text{ and } y \text{ are multiplied by } b\text{)}$$

$$a + b(x + y) = ax + ay + bx + by \text{ (} x \text{ and } y \text{ are multiplied by } (a+b)\text{)}$$

$$\begin{aligned} (a + b)^2 &= (a + b)(a + b) = (a \times a) + (a \times b) + (b \times a) + (b \times b) \\ &= a^2 + ab + ab + b^2 = a^2 + 2ab + b^2 \end{aligned}$$

When factorising, examine each term in order to look for common factors. the common factors of **a^2b and $-2ab^2$** are **a and b** (they appear in both),

$$\text{hence } a^2b - 2ab^2 \text{ can be written } (ab)(a - 2b)$$

(ab) and **$(a - 2b)$** are both factors of the complete expression **a^2b and $-2ab^2$** .

$$\begin{aligned} &ax + bx + ay + by \\ &= x(a + b) + y(a + b) = (x + y)(a + b) \\ \text{or } &= a(x + y) + b(x + y) = (x + y)(a + b) \end{aligned}$$

Algebra can be extended to include fractions.

e.g. $\frac{a}{b} + \frac{c}{d} = \frac{ad + cb}{bd}$

(bd is the LCD, ad + cb is the numerator).

EQUATIONS

The statement $a - 4 = 5$ is an equation. What we are saying is that an unknown quantity minus 4 equals 5. It does not take a genius to work out that the unknown quantity in this case is 9, there is only one value that will be correct.

The value of 'a' can be calculated using guesswork or elimination. The process of establishing that $a = 9$ is called solving the equation.

Solving Linear Equations

A linear equation is one containing only the first power of the unknown quantity.

$$5y - 5 = 3y + 9 \quad \text{or} \quad 5(m - 2) = 15$$

These are both linear equations.

When we solve linear equations, the appearance of the equation may change. For example, the first equation could be re-written as $5y - 3y = 9 + 5$ and the second as $5m - 10 = 15$. Both of these look different from the original form, but equality has been maintained and they are therefore the same.

The general rule for all equations is:

Whatever you do to one side of the equation, you must do the same to the other side.

By convention we name each side of the equation **Left Hand Side (LHS)** or **Right Hand Side (RHS)**

Equations Requiring Multiplication or Division

Example

Solve the equation

$$\frac{x}{5} = 4$$

If we multiply both sides by **5** we get:

$$\frac{x}{5} \times 5 = 4 \times 5$$

So the solution is **x = 20**

Example

Solve the equation

$$4b = 20$$

Dividing both sides by **4** we get:

$$\frac{4b}{4} = \frac{20}{4}$$

So the solution is **b = 5**

Equations Requiring Addition or Subtraction

The simplest type of linear equation is of this type:

$$x - 6 = 9$$

To solve all equations we must manipulate the equation to get the unknown on one side and the known values on the other side. In this case we must eliminate the value of **-6** from the **LHS**.

This can be done by adding **6** to the **LHS**, but we must also add **6** to the **RHS**. So the equation becomes:

$$x - 6 + 6 = 9 + 6$$

We then simplify the equation to obtain:

$$x = 9 + 6 = 15$$

So the solution is **x = 15**

A simpler way of solving this type of equation is to switch values from one side to another. When we do this, we must, however change the sign.

Example

Solve $y + 4 = 14$

If we switch the **+ 4** to the **RHS** and change the sign it becomes

$$Y = 14 - 4$$

So the solution is **y = 10**

If the equation has multiples of the unknown quantity, such as solve:

$$\text{Solve } 5x - 12 = 3$$

the first stage is the same, i.e.

$$5x = 3 + 12$$

$$\text{So } 5x = 15$$

It seems obvious that $x = 3$, but how mathematically is this achieved? If we divide both sides by **5** we will get the solution $x = 3$.

Equations Containing Unknowns on both Sides

In equations of this type we should group the unknown quantities on one side and the other terms on the other side.

For example, solve :

$$8y + 4 = 5y + 22$$

If we **subtract 4** from both sides, and also **subtract 5y** from both sides we will get:

$$3y = 18$$

The solution can then be obtained by dividing each side by **3**.

$$\frac{3y}{3} = \frac{18}{3} = y = 6$$

Note:

As in all cases of solving equations, we can and should check if our solution is correct by substituting the solution in the original equation.

$$\text{LHS } (8 \times 6) + 4 = 48 + 4 = 52$$

$$\text{RHS } (5 \times 6) + 22 = 30 + 22 = 52$$

Equations Containing Brackets

The first step is to remove the brackets and then solve as normal:

$$3(2y + 3) = 21$$

first expand the brackets to obtain

$$6y + 9 = 21$$

then subtract 9 from both sides

$$6y = 12$$

then divide both sides by 6 the solution is

$$y = 2$$

To check the solution is correct, we substitute $y = 2$ in the original equation.

LHS	$3(2 \times 2 + 3)$	$=$	$3(4 + 3)$	$=$	3×7	$=$	21
RHS	$= 21$						

Equations Containing Fractions

In this case we must multiply each term by the **LCM** of the denominators.

Example 1

$$\frac{y}{4} + \frac{3}{5} = \frac{3y}{2} - 2$$

The **LCM** of the denominators **4, 5 and 2** is **20**, so we must multiply each term in the equation by **20**

$$\frac{y}{4} \times 20 + \frac{3}{5} \times 20 = \frac{3y}{2} \times 20 - 2 \times 20$$

$$5y + 12 = 30y - 40$$

$$5y - 30y = -40 - 12$$

$$\text{so } -25y = -52$$

Note in this case we have negative values on both sides. If we swap them around and change the signs i.e. swap the **LHS** for the **RHS**

Note this is exactly the same as **25y = 52**. This can be proved by taking the equation - **25y = - 52** and adding **25y** to both sides, and then adding **52** to both sides

Example 2

Solve the equation:

$$\frac{x-4}{3} - \frac{2x-1}{2} = 4$$

The **LCM** of 3 and 2 is 6, so we multiply all of the terms by 6

$$\frac{x-4}{3} \times 6 - \frac{2x-1}{2} \times 6 = 4 \times 6$$

$$2(x-4) - 3(2x-1) = 24$$

$$2x - 8 - 6x + 3 = 24$$

$$-4x - 5 = 24$$

$$\text{so } -4x = 29$$

And the solution is:

$$x = -\frac{29}{4} = -7.25$$

TRANSPOSITION IN EQUATIONS

Consider a formula (equation) given in a certain form.

$$6a + 11 = 25 - a$$

This contains one algebraic quantity, "**a**", within an equation. Think of an equation as a statement of '**balance**'. In this one, **6a + 11** on the **LHS** equals, or balances, **25 - a** on the **RHS**.

As we have one equation and one unknown '**a**', there is only one numerical value which can produce a balance. What is it?

By manipulating (transposing is the word) the equation, it is possible to isolate the '**a**' on the **LHS** and balance it with an actual number on the **RHS**. This will then be the unique value of '**a**'. Look again at the equation.

$$6a + 11 = 25 - a$$

To remove the '**a**' on the **RHS**, we must **add 'a'** to both sides.

$$6a + 11 + a = 25 - a + a \text{ therefore } 7a + 11 = 25$$

To remove + 11, we must subtract 11 from both sides

$$7a + 11 - 11 = 25 - 11$$

$$\text{so } 7a = 14$$

$$\text{and if } 7a = 14 \text{ then } a = 2$$

We have found that **a = 2**.

This is the unique value which satisfies **6a + 11 = 25 - a**.

Study it again to see how we worked to isolate the required term 'a' on one side, and remember, what you do to one side of an equation, you must do to the other side if the balance is to be maintained.

Example 1

Here is another formula involving several algebraic symbols.

$$\text{Find } N, \quad \text{if } C = \frac{N - n}{2p}$$

Remember, we want **N** on one side by itself. It is important to get a 'feel' for the form of the equation. To help, we will put brackets around **(N - n)**. So:

$$C = \frac{(N - n)}{2p}$$

To remove the **2p** we must multiply both sides by **2p**.

$$C \times 2p = \frac{(N - n)}{2p} \times 2p$$

To remove the **-n**, we must **add n** to both sides

$$2Cp + n = (N - n) + n = N$$

That is,

$$N = 2Cp + n$$

Example 2

The volume of a cone:

$$V = \frac{\pi r^2 h}{3}$$

Find **r (the radius)**, step by step.

First multiply both sides by **3**

$$V \times 3 = \frac{\pi r^2 h}{\cancel{3}} \times \cancel{3}$$

Then divide both sides by **πh**

$$\frac{3V}{\pi h} = \frac{\cancel{\pi} r^2 \cancel{h}}{\cancel{\pi} \cancel{h}} = r^2$$

Remember, to find **r**, take the square root of **r^2** and do the same to both sides.

$$\sqrt{\frac{3V}{\pi h}} = \sqrt{r^2} = r$$

$$r = \sqrt{\frac{3V}{\pi h}}$$

This is what transposition is all about. We are re-arranging formulas expressed as equations, which then allows us to find a particular numerical value for one (unknown) quantity if the other numerical values are given.

One important point, it is only possible to find an unknown quantity if all the other values are known. This is known as '**solving an equation**'.

The rule is,

***One unknown quantity can be deduced from one equation
 Two unknowns require two different equations
 Three unknowns required three different equations, and so on.***

Construction of Equations

As already stated, maths serves as a "tool" for Engineers at the design stage. Design is the creation of a component or mechanism on paper, i.e. before it takes shape in metal or plastic. The design engineer hopefully makes it strong enough - his knowledge of materials and their strengths allow him to do this by calculation. He uses formulas and equations.

To do this, he must allocate letters to represent some variable or known quantity. He can then construct a formula or equation by using the letters within some 'reasonable' statement about the situation. He studies the situation and then makes the statement.

How do we construct equations from the facts contained within a scenario?

Example 1

Think of a number, double it, **add 6** and divide the result by **3**. What is the answer?

Let the number you think of be **A**. Doubling this number gives **2A**.

If **6** is then added, we have **2A + 6**, which must then be divided by **3**, making the

answer = $\frac{2A + 6}{3}$.

This formula can be used to calculate the answer no matter what number you think of.

Example 2

If one side of a rectangular field is twice as long as the other, and the short side is **100m**. Calculate the area of the field.

Let the short side of the field be **L**. The long side is therefore **2 x L** or **2L**. To calculate the area we multiply one side by the other, so:

$$\begin{aligned} \text{Area} &= 2L \times L = 2L^2 \quad \text{where } L \text{ equals } 100\text{m} \\ \text{Area} &= 2(100)^2 = 20000\text{m}^2 \end{aligned}$$

Example 3

A certain type of motor car cost seven times as much as a certain make of motor cycle. If two cars and three motor cycles cost **£8500**, find the cost of each vehicle.

Let the cost of a car be **C** (at present **C** is an unknown).

Let the cost of a motor cycle be **M** (another unknown).

We know that **2C + 3M = £8500** (this has two unknowns within one equation).

But we also know that **C = 7 x M**, therefore, we can substitute for **C** in the first equation.

$$\begin{aligned} 2(7M) + 3M &= £8500 \\ 14M + 3M &= 17M = £8500 \\ M &= \frac{£8500}{17} = £500 \end{aligned}$$

The cost of a motor cycle is therefore **£500**, and the cost of a car must be **7 X £500 = £3500**.

Here 2 equations were constructed from the facts, and then combined to allow a solution to be found.

In the next example, we form equations from the facts, and then transpose to produce a solution.

Example 4

Three electric radiators and five convector heaters together cost **£740**. A convector cost **£20** more than a radiator. Find the cost of each."

Let **R** represent the cost of a radiator, and **C** represent the cost of a convector.
Then

$$3R + 5C = £740$$

$$C = R + 20$$

$$\therefore 3R + 5(R + 20) = 3R + 5R + 100 = 740$$

$$\therefore 8R = 740 - 100 = 640$$

$$R = \frac{640}{8} = £80 \text{ (the cost of a radiator)}$$

$$\text{and } C = 80 + 20 = £100 \text{ (the cost of a convector)}$$

SIMULTANEOUS EQUATIONS

Consider the equation $4x - 3y = 1$. There are **2** unknowns (**x and y**) in one equation, and so the equation cannot be solved to give a single value for **x and y**. There are an infinite number of values of **x** for which there are corresponding values of **y**. For example if:

$$\begin{aligned}\text{If } x &= 4, \text{ then } y = 5 \\ \text{If } x &= 7, \text{ then } y = 9 \\ \text{If } x &= 1, \text{ then } y = 1\end{aligned}$$

However, if a second equation exists, for example $x + 3y = 19$, then these two equations can be evaluated simultaneously to give single values for **x and y**.

The process is simple and involves modifying the equations, whilst still preserving the equalities.

$$\begin{aligned}4x - 3y &= 1 & (1) \\ x + 3y &= 19 & (2)\end{aligned}$$

The method of solution of all simultaneous equations is to:

- **First manipulate one or both of the equations so that the coefficient of one of the unknowns is the same in both equations.**
- **Then add or subtract one of the equations from the other to produce a third equation with only one unknown. The other having become zero.**
- **Solve the new equation to find the unknown.**
- **Put the solution into one of the original equations to find the other unknown.**
- **Put both solutions into the equation not used in the stage above to check your answers.**

Example

Using these two equations as an example:

$$4x - 3y = 1 \quad (1)$$

$$x + 3y = 19 \quad (2)$$

We do not need to manipulate either of the equations because the co-efficient of **y** is the same in both equations. Therefore, we can eliminate the “**y**” value simply by adding the two equations. The result is:

$$5x = 20 \quad \text{So } x = 4$$

If we then substitute **x = 4** in the second equation we get:

$$4 + 3y = 19 \quad \text{So } 3y = 19 - 4 = 15 \quad \text{So } y = 5$$

Our solutions are **x = 4** and **y = 5**

Example 1

$$2x + 3y = 8 \quad (1)$$

$$3x + 5y = 11 \quad (2)$$

Multiply equation (1) by the coefficient of x in equation (2).

$$(2x + 3y = 8) \times 3 = 6x + 9y = 24$$

Multiply equation (2) by the coefficient of x in equation (1).

$$(2x + 5y = 8) \times 2 = 6x + 10y = 22$$

$$\text{So } 6x + 9y = 24 \quad (3)$$

$$6x + 10y = 22 \quad (4)$$

Subtract equation (4) from (3)

$$0x - 1y = 2.$$

$$\text{so } -y = 2 \quad \text{and} \quad y = -2$$

substitute $y = -2$. in either equation (1) or (2) to solve for x . I have selected (1).

$$2x + 3(-2) = 8 \quad \text{therefore} \quad 2x = 14 \quad \text{and} \quad x = 7$$

Check your answer by substituting both values in equation (2). Do not use equation (1) because it will not highlight an error. If you had used equation (2) to find x , then the check should be carried using equation (1).

$$3x + 5y = 11$$

$$3(7) + 5(-2) = 11 \quad \text{therefore} \quad 21 + (-10) = 11 - \text{correct}$$

The same result would be found if y was eliminated as shown below.

$(2x + 3y = 8) \times 5$	$10x + 15y = 40 \quad (3)$
$(3x + 5y = 11) \times 3$	$9x + 15y = 33 \quad (4)$

$$x = 7 \text{ etc.}$$

QUADRATIC EQUATIONS

Any equation of the form $y = ax^2 + bx + c$, where **a**, **b** and **c** are numbers, is known as a quadratic equation. An equation of this type will produce a curve called a '**Parabola**'.

The actual value for coefficients **a**, **b** and **c** will determine the exact shape and position of the curve.

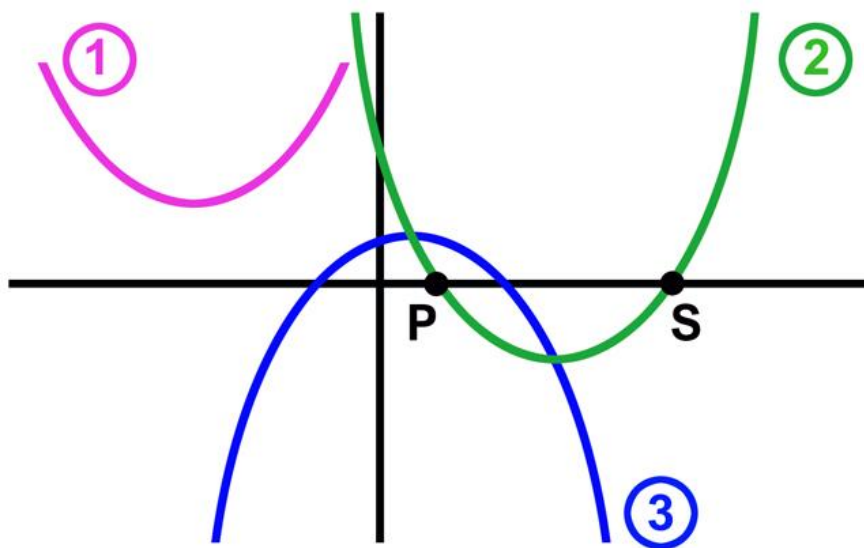


Fig 02. 1 - Parabola

It will be noted that one of the curves cuts the **x-axis** at points **P** and **S**. **P** and **S** are known as the roots of the equation. Alternatively, **P** and **S** are the values of **x** which satisfy the condition $y = ax^2 + bx + c = 0$.

It can be shown that the Roots are found to be equal to:

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

This equation gives two values, one for **P** the other for **S**.

Example

Find the roots of

$$y = 6x^2 - 5x - 6 \quad (a = 6, b = -5, c = -6)$$

$$x = \frac{-(-5) \pm \sqrt{(-5)^2 - 4(6)(-6)}}{2(6)}$$

$$x = \frac{5 \pm \sqrt{25 + 144}}{12} = \frac{5 \pm 13}{12}$$

$$x = \frac{18}{12} \text{ or } \frac{-8}{12} = 1\frac{1}{2} \text{ or } -\frac{2}{3}$$

Note

Depending on a, b and c, it is possible that $b^2 - 4ac$ results in a negative value.

It has been considered impossible to find the square root of a negative value. The equation concerned is then said to have no real roots. When $b^2 - 4ac$ is negative, the equation is said to have complex roots, where the roots comprise both a real and imaginary component. This concept is not considered in these notes.

LIST OF ILLUSTRATIONS

Fig 02. 1 - Parabola..... 22

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$$3 = 9^{\frac{1}{2}} \quad \text{or} \quad \sqrt{9}$$

$$2 = 32^{\frac{1}{5}} \quad \text{or} \quad \sqrt[5]{32}$$

$$4 = 64^{\frac{1}{3}} \quad \text{or} \quad \sqrt[3]{64}$$

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SECTION 03

NUMBERS

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NUMBERS

Indices and Powers

It is often necessary to multiply a number by itself once, twice or several times. To indicate this, a method of notation has evolved, which is both convenient and capable of being extended to introduce other concepts.

3×3 is written as 3^2
 $2 \times 2 \times 2 \times 2 \times 2$ is written as 2^5
 $4 \times 4 \times 4$ is written as 4^3 etc, etc.

In the above examples, the number being multiplied by itself is known as the '**Base**' and the number of times it is multiplied by itself is known as the '**Power**' or '**Index**'. Alternatively, the number **2** has been raised to **power 5**.

Power 2 is generally referred to as the '**Square**'

Power 3 is generally referred to as the '**Cube**'.

$3 \times 3 = 3^2 = 9$
9 is the square of 3 or 3 squared equals 9
 $4 \times 4 \times 4 = 4^3 = 64$
64 is the "cube" of 4 or 4 cubed equals 64

But put another way:

- **3 is said to be the square root of 9**
- **4 is the cube root of 64**
- **2 is the fifth root of 32.**

The method of notation used is that:

$$\begin{aligned} 3 &= 9^{\frac{1}{2}} & \text{or} & \sqrt{9} \\ 2 &= 32^{\frac{1}{5}} & \text{or} & \sqrt[5]{32} \\ 4 &= 64^{\frac{1}{3}} & \text{or} & \sqrt[3]{64} \end{aligned}$$

It is possible to re-write the above, so that $3 = 9^{0.5}$, $2 = 32^{0.2}$ and $4 = 64^{0.333}$. Where the power is expressed as a decimal, instead of a fraction.

To allow the use of numbers involving powers and indices, some rules have evolved, which are reproduced, using the symbol **N** to represent any base number.

Rule 1:

$$\begin{aligned} N^2 \times N^3 &= N^5 \\ N^a \times N^b &= N^{(a+b)} \end{aligned}$$

Rule 2:

$$N^5 \div N^2 = N^3 = \frac{N \times N \times N \times N \times N}{N \times N \times N} = N^2$$

$$N^a \div N^b = N^{(a-b)}$$

Rule 3:

$$(N^2)^3 = N^2 \times N^2 \times N^2$$

using rule 1 this equals N^6

$$(N^a)^b = N^{(a \times b)} \text{ or } N^{ab}$$

Rule 4

$$N^2 \div N^2 = N^{(2-2)} = N^0$$

Any number divided by itself equals **1** so $N^0 = 1$

Therefore:

$$\frac{1}{N^2} = N^0 \div N^2$$

using **rule 2** this equals N^{-2}

$$\therefore \frac{1}{N^a} = N^{-a} \quad \text{also} \quad \frac{1}{N^{-a}} = N^a$$

Because

$$1 \div N^2 \text{ is the same as } N^0 \div N^2 = N^{(0-2)} = N^{-2}$$

Rule 5

If

$$N^{1/3} \times N^{1/3} \times N^{1/3} = N^1 = N$$

then $N^{1/3}$ must be the third root of **N**, because the only number that can be multiplied by itself **3** times to make **N** is the **third root of N**.

$$\therefore N^{1/3} = \sqrt[3]{N}$$

similarly if

$$N^{2/3} \times N^{2/3} \times N^{2/3} = N^2$$

then $N^{2/3}$ must be the third root of N^2

$$\therefore N^{2/3} = \sqrt[3]{N^2}$$

$$\text{SO } N_{a/b} = \sqrt[b]{N^a}$$

STANDARD FORM

If the number **8.347** is multiplied by **10,000** then the product is **83470**. This calculation can be written as **$8.347 \times 10^4 = 83470$** .

When **83470** is written as **8.347×10^4** , it is known as **Standard Form**.

A number in standard form has two parts. The first part is a number between **1** and **10** (but does not equal **10**), and the second part is **10** raised to some whole number power.

- The first part is called the '**Mantissa**'.
- The second part the '**Exponent**'.

To express a number in standard form, move the decimal point left or right to create a number between **1** and **10** (the mantissa), and then create the **exponent**. The value of which equals the number of places by which the decimal point has been moved.

- If the point was moved **Left**, the power is **positive**
- If the point was moved **Right**, the power is **negative**.

Examples:

$$\begin{array}{lcl} 526 & = & 5.26 \times 10^2 \\ 0.3716 & = & 3.716 \times 10^{-1} \\ 0.002 & = & 2.0 \times 10^{-3} \end{array}$$

NUMBERING SYSTEMS

The most widely used system of numbers is the decimal system, based on the hindu-arabic symbols **0, 1, 2, 3** etc but roman symbols such as **V, X, L** and **C** are also well known and understood. To-day, the practice of engineering requires a measure of competence in handling several different systems of numerals.

In general a system of numeration consists of a set of symbols together with a rule by which the symbols can be combined together.

Number is the property associated with a set or collection of things. It is independent of the nature of the individual items in the set. The number fourteen may be written as **14** or **XIV**. In this case the number is the same but the system or numeration is different.

Decimal System of Numeration

In the decimal system, the symbols are combined by arranging them in a horizontal line, the contribution that each digit makes being governed by its position. A decimal point enables numbers less than one to be represented.

Example 1

Decimal **368** is really:

$$(3 - 10^2) + (6 - 10^1) + (8 - 1^0)$$

or in column form:

10^2 (Hundreds)	10^1 (Tens)	10^0 (Units)
3	6	8

Example 2

Decimal 452.64 is really:

$$(4 - 10^2) + (5 - 10^1) + (2 - 10^0) + (6 - 10^{-1}) + (4 - 10^{-2})$$

or in column form:

10^2	10^1	10^0	10^{-1}	10^{-2}
4	5	2	6	4

Ten is known as the '**Base**' or '**Radix**' of the decimal system. The '**Index**' indicates the power to which the base is raised.

The base, and the particular index to which it is raised is called the '**Weight**'. e.g.

least significant weight	=	10^0	=	1
next most significant weight	=	10^1	=	10

The numbers by which weight is multiplied are called '**Digits**'. In practice only the digits of the system are written, the weight being implied e.g.

368, 53.24

Note:

0 is counted as a digit, so that there are ten digits in the decimal system, 0 to 9 inclusive.

Binary System of Numeration

Only the symbols **0** and **1** are used and the base is two, otherwise the system of numeration is the same as before. The two digits **0** and **1** are referred to as bits, an abbreviation of binary digits.

Example 1

101101 is really:

$$(1 - 2^5) + (0 - 2^4) + (1 - 2^3) + (1 - 2^2) + (0 - 2^1) + (1 - 2^0)$$

or in column form:

2^5	2^4	2^3	2^2	2^1	2^0
1	0	1	1	0	1

(= 45 in decimal)

Example 2

110.11 is really:

$$(1 - 2^2) + (1 - 2^1) + (0 - 2^0) + (1 - 2^{-1}) + (1 - 2^{-2})$$

or in column form:

2^2	2^1	2^0	2^{-1}	2^{-2}
1	1	0	1	1

(= 6.75 in decimal)

Note

All digits to the right of the binary point refer to negative powers.

The binary system is very suitable for use with electrical switching circuits. A switch is either off or on corresponding, for example, to **0** and **1** respectively. There is no ambiguity.

Octal System of Numeration

In the octal system of numeration the symbols **0** to **7** are used and the **base** is **8**. Again the system of numeration is the same as that used for decimal and binary, with each column increasing by a power of one as you move from right to left.

Example 1

$$376^8 \text{ is really } (3 \cdot 8^2) + (7 \cdot 8^1) + (6 \cdot 8^0)$$

or in column form:

8^3	8^2	8^1	8^0
0	3	7	6

in decimal

$$(3 \times 64) + (7 \times 8) + (6 \times 1) = 192 + 56 + 6 = 254$$

Example 2

$$37.13^8 \text{ is really } (3 \cdot 8^1) + (7 \cdot 8^0) + (1 \cdot 8^{-1}) + (3 \cdot 8^{-2})$$

or in column form:

8^2	8^1	8^0	8^{-1}	8^{-2}
0	3	7	1	3

in decimal

$$(3 \times 8) + (7 \times 1) + (1 \times 0.125) + (3 \times 0.015625) = 31.140625$$

Note:

All digits to the right of the octal point refer to negative powers.

Conversion to Other Bases

Conversion from decimal to any other base can be achieved by dividing the decimal number repeatedly by the new base and recording the remainder. The remainder gives the number in the new base and should be read from bottom to top.

Example

Convert 29_{10} to binary:

2	29			
2	14	Rem	1	
2	7	Rem	0	
2	3	Rem	1	
2	1	Rem	1	
	0	Rem	1	

Result 11101_2

Example 2

Convert 57_{10} to octal:

8	57			
8	7	Rem	1	
	7	Rem	7	

Result 71_8

Example 3

Convert 63_{10} to hexadecimal:

16	63			
16	3	Rem	15(F)	
	0	Rem	3	

Result $3F_{16}$

To convert Binary Numbers to Decimal

The easiest way to convert from binary to decimal is to remember the weightings, or if necessary write the weightings above each binary digit, and add them up.

Example 1

Convert **1 0 1 1 0 1** to decimal.

2^5 (32)	2^4 (16)	2^3 (8)	2^2 (4)	2^1 (2)	2^0 (1)
1	0	1	1	0	1

$$(1 \times 2^5) + (0 \times 2^4) + (1 \times 2^3) + (1 \times 2^2) + (1 \times 2^1) + (1 \times 2^0) = 45_{10}$$

An alternative method for long binary numbers is to take the left-hand digit, double it and add the result to the next digit to the right as shown below (double and add to next digit to the right).

1	0	1	1	1	0
1	2	5	11	23	46

To Convert Binary to Octal or Vice Versa

Each octal digit can be represented by **3** binary digits. Therefore, to convert from binary to octal:

1. Split the binary number into groups of 3 digits starting from the right.
2. Weight the numbers in each group 4 – 2 – 1
3. Find the total of each group of 3 digits, the result is the octal value.

Example 1

Convert **1 0 1 1 1 0 0 1** to Octal

Binary No		1	0		1	1	1		0	0	1
Weighting	4	2	1		4	2	1		4	2	1
Octal No (sum)		2				7				1	

Answer **10111001₂ is equal to 271₈**

The reverse process should be used to convert octal to binary. Convert each digit into a **3** digit binary number keeping the order of digits the same. Work from the bottom to the top of the table shown above to convert **271₈** to binary.

To Convert Binary to Hexadecimal or Vice Versa

The process for converting a binary number to a hexadecimal one is the same as that used to convert binary numbers to octal. Each hexadecimal digit can be represented by 4 binary digits, therefore the binary number is split into groups of 4 digits starting from the right. The weightings this time are **8 – 4 – 2 – 1**.

Again, the reverse process is used to convert from hexadecimal to binary. Convert each hexadecimal digit into its binary equivalent keeping the order the same.

Example 1

Convert **A7**₁₆ to binary.

Hexadecimal No	A				7			
Weightings	8	4	2	1	8	4	2	1
Binary No	1	0	1	0	0	1	1	1

Answer **A7**₁₆ is equal to **10100111**₂

LOGARITHMS

Logarithms are a mathematical concept that was developed to simplify multiplication and division of large numbers. Logarithms enable multiplication and division to be performed using addition and subtraction.

The use of logarithms is no longer so widespread as the electronic calculator has become so readily available. Remembering that when, for example,

25 is written as **5²**, **5** is known as the '**Base**' and **2** as the '**Power**', then the logarithm of **25** can be expressed as **2, to the base 5**.

The general definition is, that if $y = a^x$ then $x = \log_a y$

So logarithms can be calculated for any **base a**, but generally only logarithms to the base of **10** or **e (2.71)** are used, and are commonly available in tabular form. However, logarithms are more easily obtained from the calculator.

An example of the function of logarithms is shown overleaf.

Example

Calculate

$$6,412 \times 23,162$$

From the calculator

$$\log_{10} \text{ of } 6,421 = 0.80699$$

$$\log_{10} \text{ of } 23,162 = 1.36478$$

$$\text{so } 6,412 \times 23,162$$

$$= 10^{0.80699} \times 10^{1.36478}$$

and using the laws of indices

$$6,412 \times 23,162 = 10^{0.80699 + 1.36478}$$

$$= 10^{(2.17177)}$$

It is now necessary to find the **base 10** number whose logarithm is **2.17177**.

The calculator shows this to be **148.51474** (this is the **anti-log of 2.17177**). If the calculator is used to solve **6.412 x 23.162**, the product is **148.51474**.

It is important to realise that this example shows how logarithms can be used, in practice, the calculator is used as normal. If a division is to be performed, the powers of logs are **subtracted**.

It is the concept of a logarithm that is important at this stage, because it re- appears later.

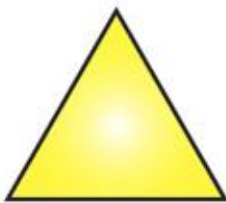
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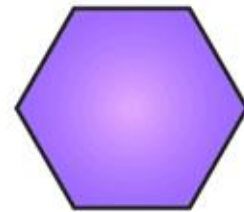
Triangle



Quadrilateral



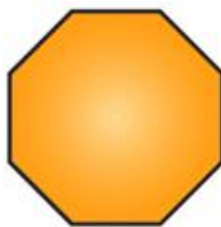
Pentagon



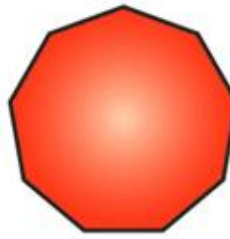
Hexagon



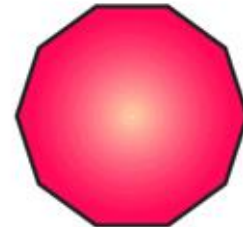
Heptagon



Octagon



Nonagon



Decagon

PART 66 MODULE 6 B1/B2

SECTION 04

GEOMETRY

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GEOMETRY

Angular Measurement

If two straight lines are drawn, we can see that they make an "angle".

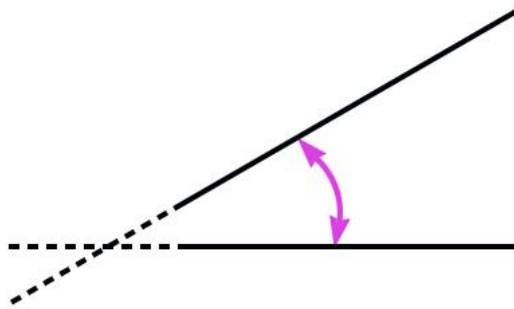


Fig 04. 1 - An Angle

But how are 'angles' expressed or measured. Consider a single line, and rotate it through a complete revolution.

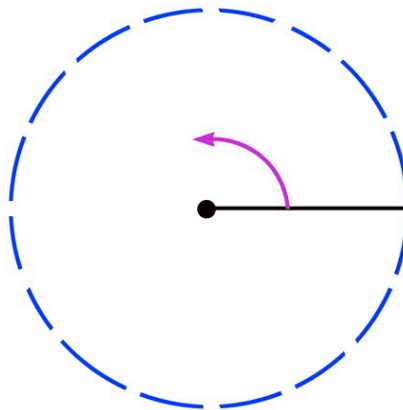


Fig 04. 2 - A Circle

Then an angle that this line has turned is **360°**.

A degree is $\frac{1}{360}$ of a revolution.

Note that half a revolution is therefore 180°

and a right angle ($\frac{1}{4}$ of a revolution) is 90°.



Fig 04. 3 - Angles

Note that 1 degree can be sub-divided into 60 minutes and 1 minute can be sub- divided into 60 seconds (very small).

A few definitions are included here:

- **An Acute angle** - **less than 90°**
- **An Obtuse angle** - **between 90° and 180°**
- **A Reflex angle** - **greater than 180°**
- **Complementary angles** - **their sum is 90°**
- **Supplementary angles** - **their sum is 180°**

ANGLES ASSOCIATED WITH PARALLEL LINES

Now consider 2 parallel lines, cut by a transversal.

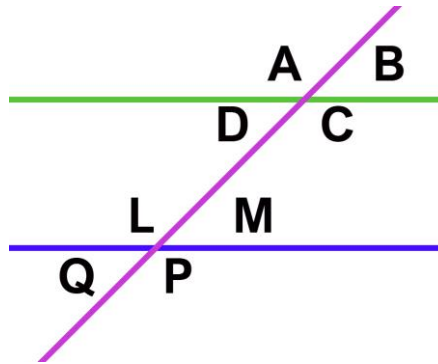
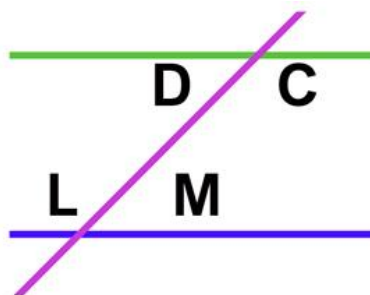


Fig 04. 4 - Parallel Lines cut

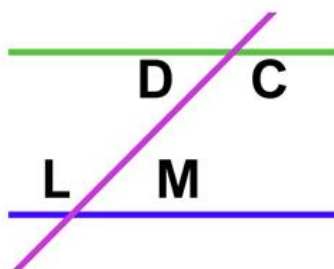
$A = C$, $B = D$ (are opposite and equal), similarly $L = P$, and $M = Q$.

Also $A = L$, $D = Q$, etc. etc. (they are corresponding angles)



$D = M$, $C = L$
 (Alternate Angles)

Fig 04. 5 - Alternate Angles



$D + L = 180$ ($= C + M$)
 (These are Interior Angles
 and are Supplementary)

Fig 04. 6 - Interior Angles

GEOMETRIC CONSTRUCTIONS

There are many different shapes associated with geometry. The more common ones are described in the following text.

Triangle

A Triangle obviously has 3 sides and 3 (internal) angles. The sides are often represented by the 3 (small) letters **a**, **b** and **c**; the angles by the (large) letters **A**, **B** and **C**.

The **3** angles add up to **180°**.

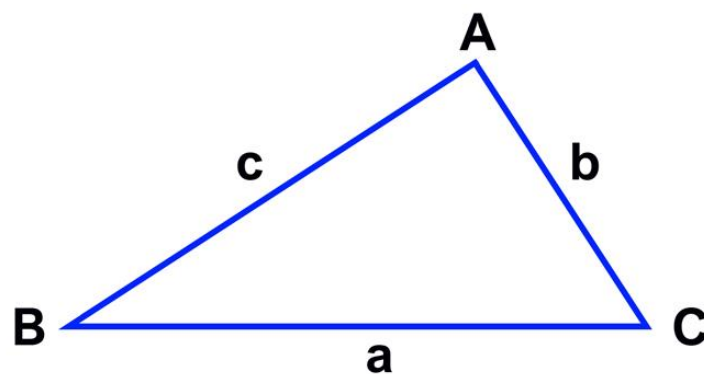


Fig 04. 7 - Triangle

The construction of a dotted line parallel to **AB** and an extension of **BC** proves this.

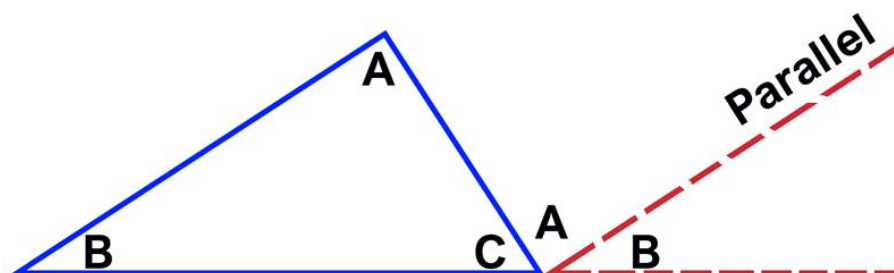


Fig 04. 8 - Extended Triangle

The area of a triangle = $\frac{1}{2}$ base x vertical height

Triangle Types

There are many different types of triangle. The main types and features are summarised as follows:

- **Acute-Angled Triangle**
 - All of its angles are less than 90° .
- **Obtuse-Angled Triangle**
 - Has one angle greater than 90° .
- **Scalene Triangle**
 - Has three sides of different lengths.
- **Right-Angled Triangle**
 - One of its angles equal is to 90° .
 - The longest side is opposite the 90° angle (right-angle) and is called the hypotenuse.
- **Isosceles Triangle**
 - Two sides and two angles are equal.
 - The equal angles lie opposite to the equal sides.
- **Equilateral Triangle**
 - All its sides and angles are equal.

Similar & Congruent Triangles

You may study two triangular shapes and estimate whether they are the same or not. We need to be more precise.

If they have the same shape, we are really saying that their angles are the same, they are then described as similar triangles. Similar triangles do not have to be the same size. One triangle may have sides twice or ten times as large as another triangle and still be classified as similar.

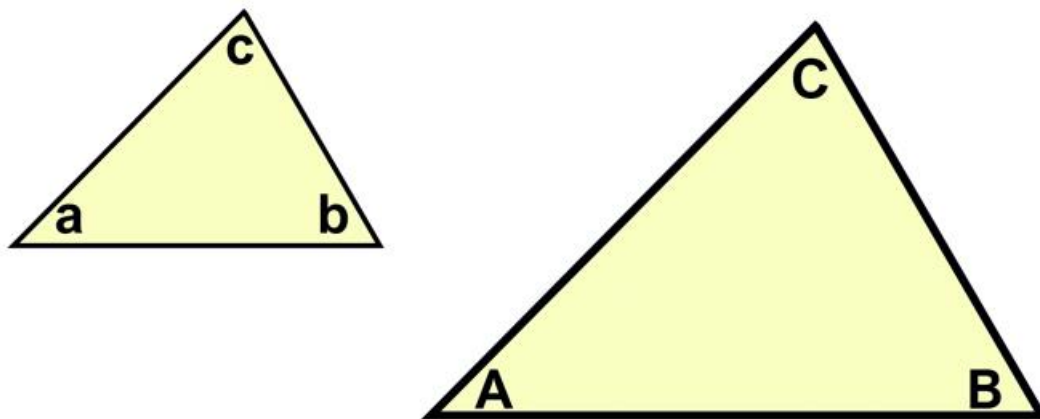


Fig 04. 9 - Similar Triangles

If they are exactly the same shape and size, their sides are the same length, then they are described as **Congruent triangles**.

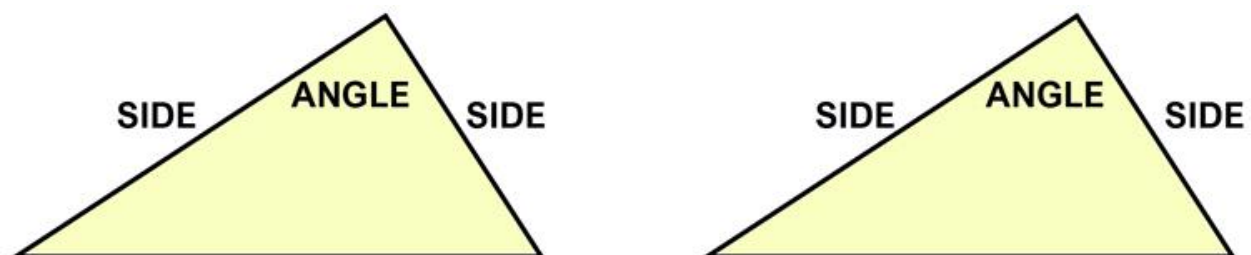


Fig 04. 10 - Congruent Triangles

It is sometimes necessary to determine whether triangles are **Congruent**. A simple criteria exists to assist us.

Two triangles are congruent if:

- **Their corresponding sides are of equal length.**
 - **(side, side, side)**
- **They have two angles and the common side equal.**
 - **(angle, side, angle)**
- **They have two sides and the included angle is equal.**
 - **(side, angle, side)**
- **The hypotenuse and one side of a right-angled triangle are equal to the hypotenuse and the corresponding side of another right-angled triangle.**

POLYGON

A Polygon is a geometric closed figure bounded by straight lines. The term poly means multi. A triangle has the least number of sides. Other multi-sided figures have names indicating the number of sides. Hence:

- **Pentagon – 5 Sided**
- **Hexagon – 6 Sided**
- **Octagon – 8 Sided**

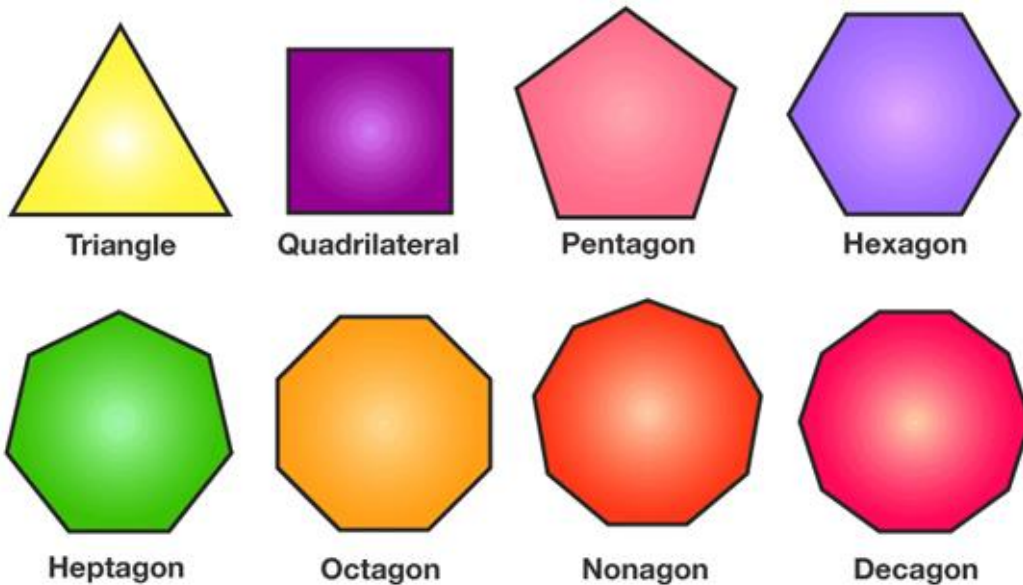


Fig 04. 11 -Polygons

Quadrilaterals

A Quadrilateral is any four-sided shape. There are various types, some are common and you are probably familiar with their names. Some are not so common.

Since a quadrilateral has four sides, it can be divided into two triangles. The sum of its angles must therefore be 360° .

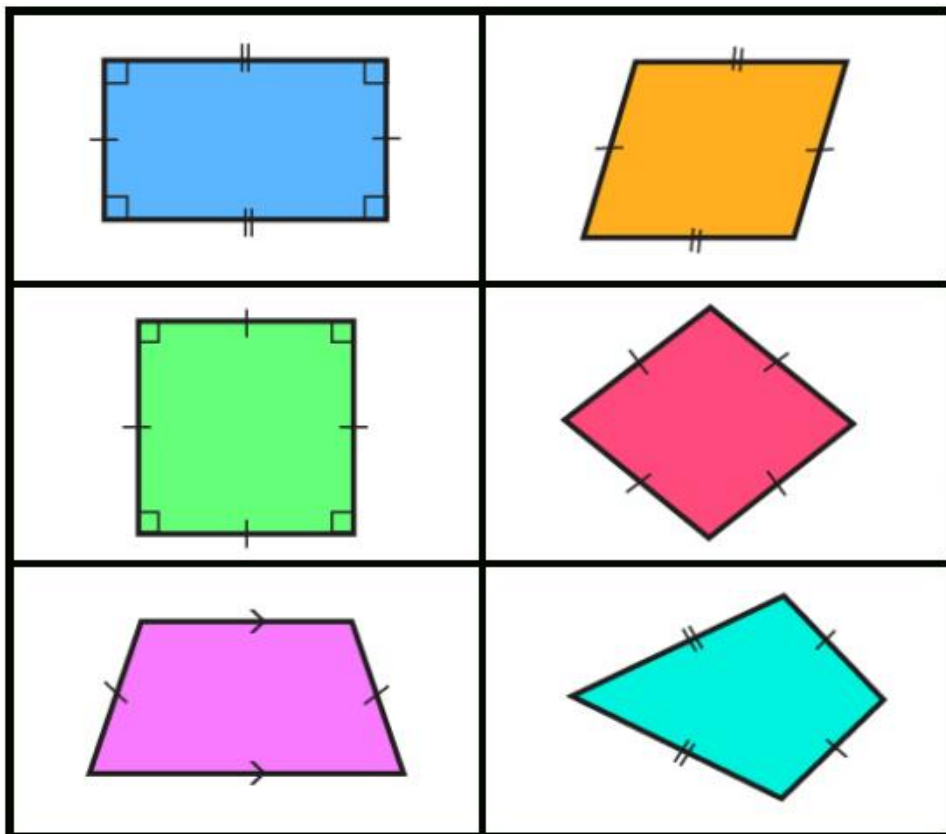


Fig 04. 12 -Quadrilaterals

Parallelogram

A Parallelogram has both pairs of opposite sides parallel. The following properties apply to parallelograms:

- **Each pair of opposite sides is equal in length.**
- **Each pair of opposite angles are equal.**
- **The diagonals bisect each other.**
- **The diagonals bisect the parallelogram and form two congruent triangles.**

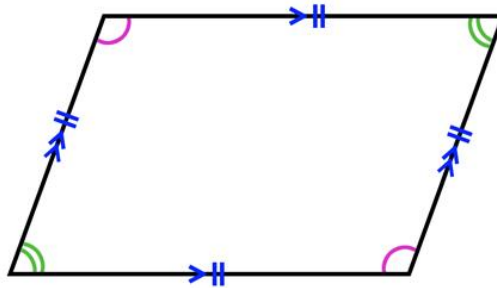


Fig 04. 13 -Parallelogram

Rectangle

A Rectangle is a parallelogram with its angle equal to 90° . It has the same properties as a parallelogram with the addition that the diagonals are equal in length.

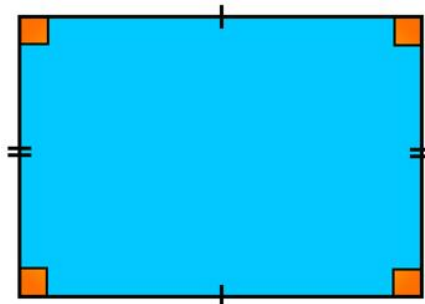


Fig 04. 14 -Rectangle

Rhombus

A Rhombus is a parallelogram with all of its sides equal in length. It also has all of the properties of a parallelogram and the following additional properties:

- **The diagonals bisect at right angles**

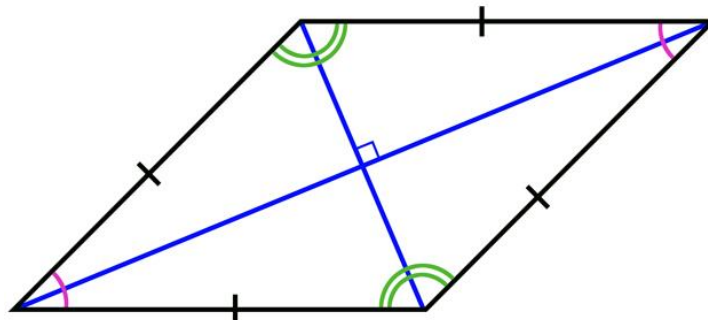


Fig 04. 15 -A Rhombus

Square

A Square is a rectangle with all the sides equal in length. It has all the properties of a parallelogram, rectangle and rhombus.

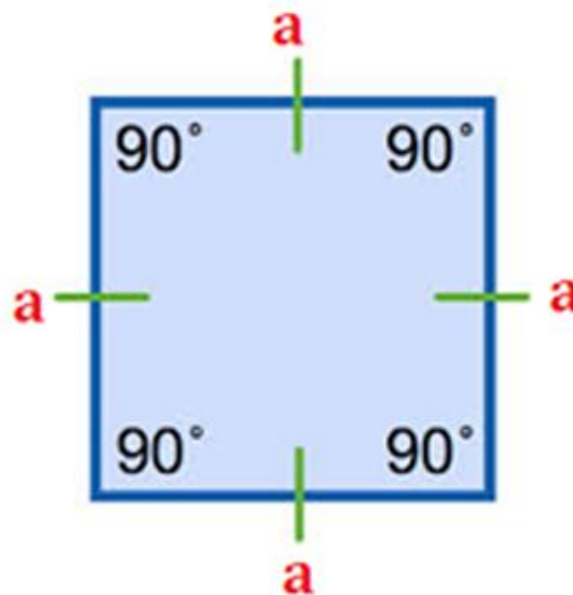


Fig 04. 16 -A Square

Trapezium

A Trapezium is a quadrilateral with one pair of sides parallel.

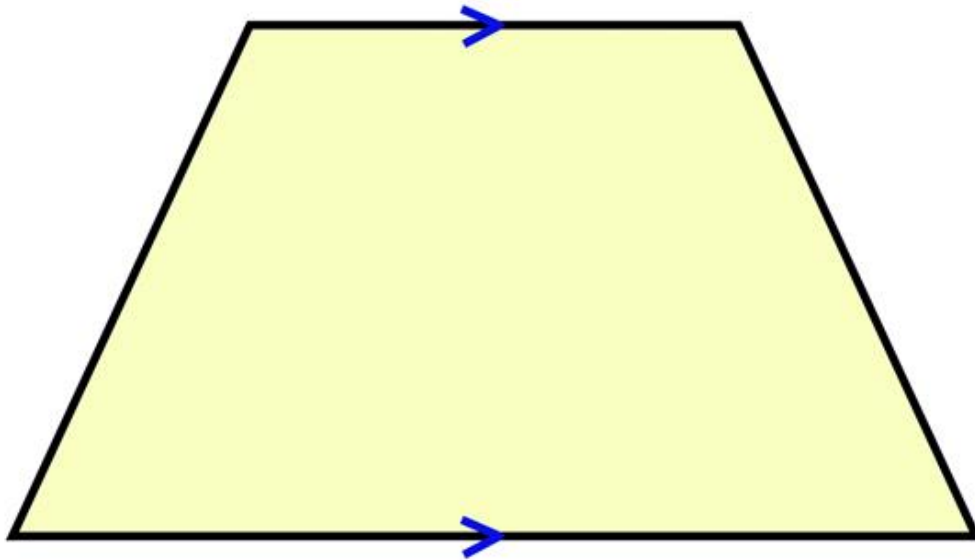


Fig 04. 17 -A Trapezium

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Circles

Circles are not just particular mathematical shapes but are involved in our everyday life, for example, wheels are circles, gears are basically circular and shafts revolve in a circular fashion. Hence, we must be aware of some important definitions and properties.

If the line **OP** is fixed at **O** and rotated around **O**, the point **P** traces a path which is circular - it forms a circle.

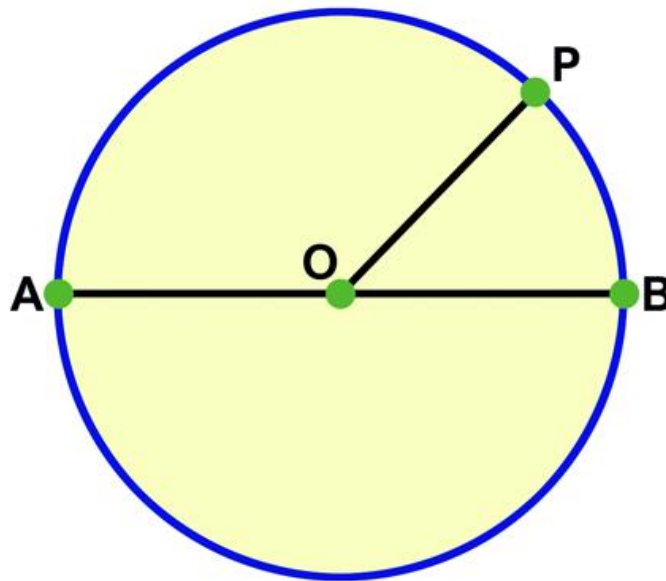


Fig 04. 18 - Circle and Diameter

The length **OP** is the Radius of the circle. Note that **OP = OA = OB** and that the length of the line **AB** is clearly equal to twice the radius. **AB = 2OP**. **AB** is the Diameter of the circle (**D = 2R**).

We already know that if **OP** is rotated through 1 complete revolution, it will have rotated through **360 degrees**, but what is the distance travelled by **P** in tracing this circular path?

Put another way, how far will a wheel whose radius is **R**, roll along a surface, during one revolution?

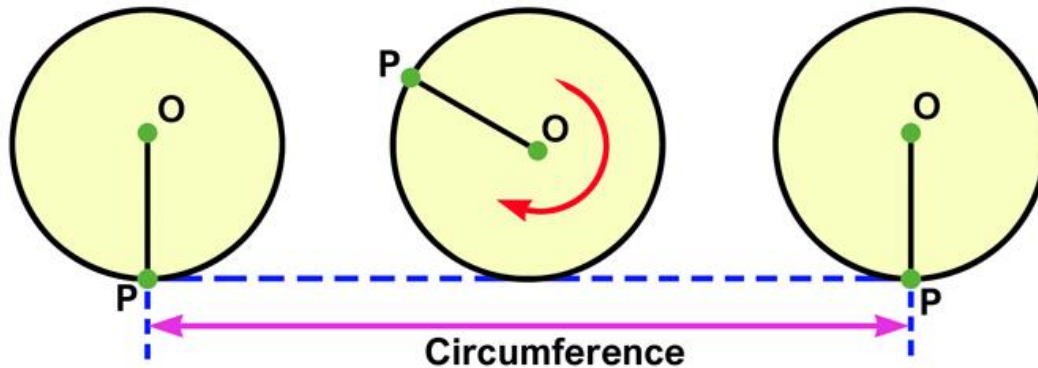


Fig 04. 19 - Circumference

The distance, known as the **Circumference** is obviously dependent on the length of the length of the diameter, but can be calculated precisely from the equation:

$$C = \pi D (= 2\pi R)$$

The value π is actually the ratio between the circumference of a circle and its diameter.

π (Greek letter, pronounced "pi") can be approximated to **3.142**.

It will certainly be found on a scientific calculator, but the fraction $\frac{22}{7}$ is a very good approximation.

Tangent

The line **AP** drawn so that it touches the circle at **point P** is known as the '**Tangent**' to the circle.

It should be noted that **AP** is always at right-angles to the **radius OP**.

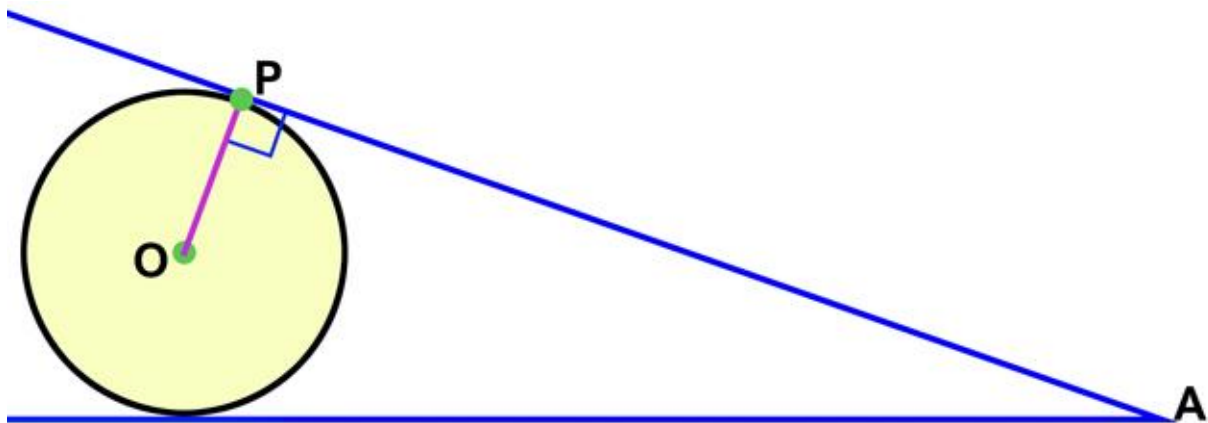


Fig 04. 20 -A Tangent

Example:

A wheel, diameter **715 mm**, makes **30** revolutions. How far does it move from its start point?

The distance moved in 1 rev. = the length of the circumference.

$$\therefore \text{Distance in 1 Rev} = \pi \times \text{Diameter}$$

$$= (\pi)(715)\text{mm}$$

$$\therefore \text{Distance in 30 Revs} = (30)(\pi)(715)$$

$$= 67410\text{mm}$$

$$= 67.4 \text{ meters}$$

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Radian Measure

We already know that an angle of **360°** represents 1 complete revolution. But there is another important unit of angular measurement, known as the **Radian**.

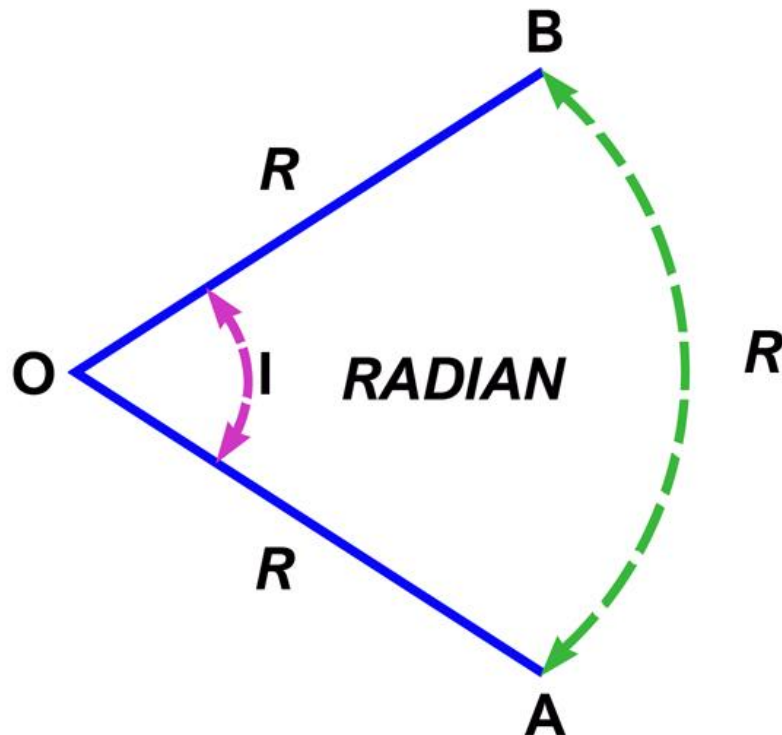


Fig 04. 21 -A Radian

Consider a circle of **radius R** and consider an **arc AB**, where length is also equal to **R**. The angle at the centre of the circle, **AOB** is then equal to **1 Radian**.

It can be deduced that **1 revolution** is equivalent to **2π Radians**, i.e. **1 rev = 6.2832 rads**. Therefore **360° = 2π rads**, and we can derive conversion factors, as that;

$$1^\circ = \pi/180 \text{ radians, or } 180^\circ/\pi = 1 \text{radian (approx } 57.3^\circ)$$

One final and useful point concerning radian measure.

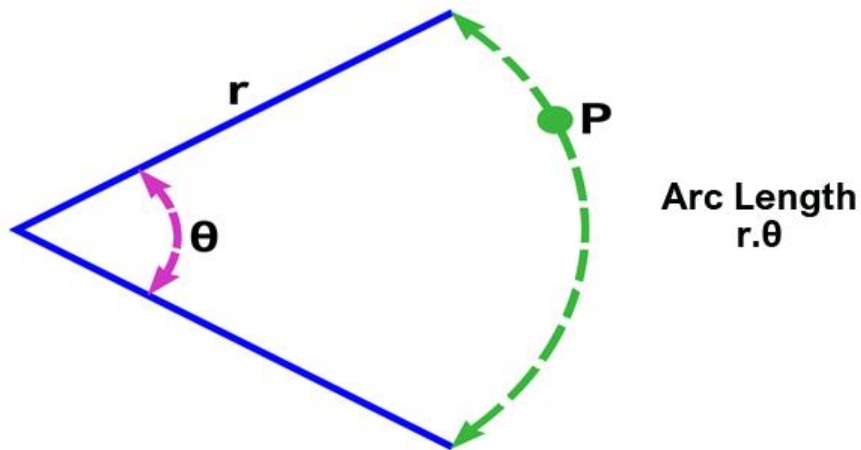


Fig 04. 22 - Arc Length

If an arc of a circle, **radius r**, subtends an angle, equal to **θ Radians**, the length of the arc is **$r.\theta$** .

Note also that if a point **P** is moving with **speed N**, then the rotational speed **ω** is equal to:

$$\frac{N}{r} \quad (N = r.\omega)$$

ω is expressed in **Radians per second**.

AREA AND VOLUME

Area

We are already familiar with the concept of length, e.g. the distance between 2 points, we express length in some chosen unit, e.g. in meters. If we want to fit a picture-rail along a wall, all we need to know is the length of the wall, so that we can order sufficient rail. But if we wish to fit a carpet to the room floor, the length of the room is insufficient. Obviously we also need to know the width. This two-dimensional concept of size is termed **Area**.

Rectangular Area

Consider a room **4m by 3m** as shown. Clearly it can be divided up into **12** equal squares, each measuring **1m by 1m**. Each square has an area of **1 square meter**. Hence, the total area is **12 square meters** (usually written as **12m²** for convenience).

So, to calculate the area of a rectangle, multiply length of one side by the length of the other side.

4m x 3m = 12m² (Don't forget the m²).

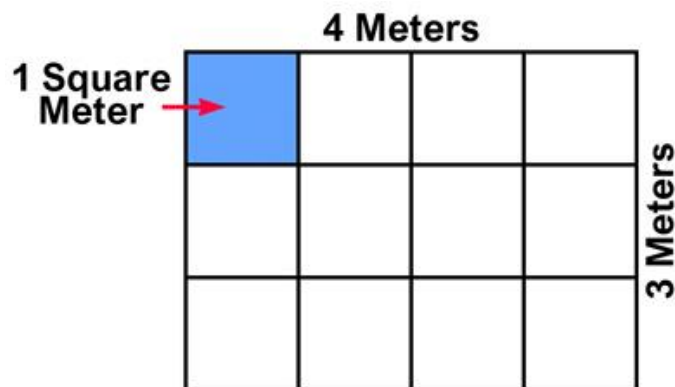


Fig 04. 23 - Square Meter

Area of Triangles

This concept can be extended to include non-rectangular shapes. Consider the triangles **ABC** and **ADC** which together form a rectangle **ABCD**.

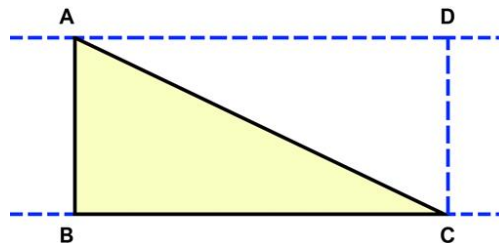


Fig 04. 24 -Area of Triangle

Inspection reveals the 2 triangles are congruent. Hence their areas are equal and the area of **ABC** = $\frac{1}{2}$ area of **ABCD**.

If we consider this diagram, the area of the triangle can be seen to equal:

$$\frac{1}{2} \times \text{base} \times \text{perpendicular height.}$$

This is true for any triangle but remember, it is the '**Perpendicular Height**'.

Note again that base (in meters) x height (in meters) gives m^2 .

A theorem exists stating that triangles with the same base and drawn between the same parallels will have the same area.

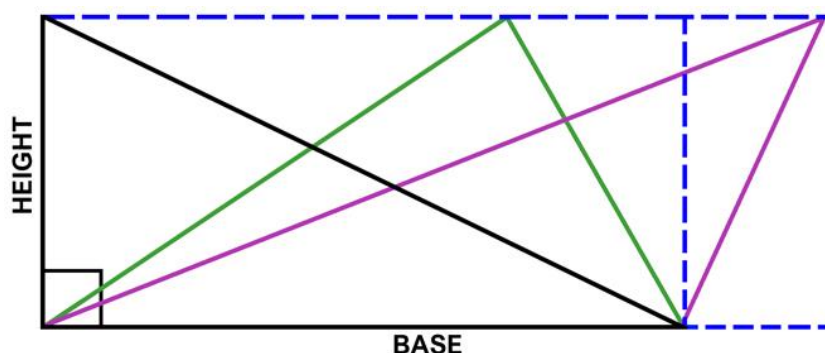


Fig 04. 25 -Same Areas

Area of Circular Shapes

The area of a circle is given by the formula:

$$A = \pi r^2 \text{ (where } r = \text{radius) or}$$

$$A = \pi \left(\frac{d}{2}\right)^2 = \frac{\pi d^2}{4} \text{ (if the diameter is given } r = \frac{d}{2}$$

Remember that any area is so many square units. So the area of a circle must include a '**squared**' term;

Example: What is the area of a semi-circle where the diameter is **30cm**?

$$\text{Area of a circle} = \pi \left(\frac{30}{2}\right)^2$$

$$\begin{aligned} \text{Semi-Circle} &= \left(\frac{1}{2}\right) (\pi) \left(\frac{30}{2}\right)^2 \\ &= \left(\frac{1}{2}\right) (\pi) (15)^2 = 353.43\text{cm} \end{aligned}$$

Area of Other Shapes

The table below indicates the areas of many common shapes.

SHAPE	AREA
CIRCLE	πR^2
TRIANGLE	$\frac{1}{2} \text{ Base x Height}$
RECTANGLE	Base x Height
SQUARE	Side^2
PARALLELOGRAM	$\text{Base x Vertical Height}$
TRAPZIUM	$\frac{1}{2}(\text{Sum of length of parallel sides}) \times \text{Vertical Height}$

Fig 04. 26 - Area of Other Shapes

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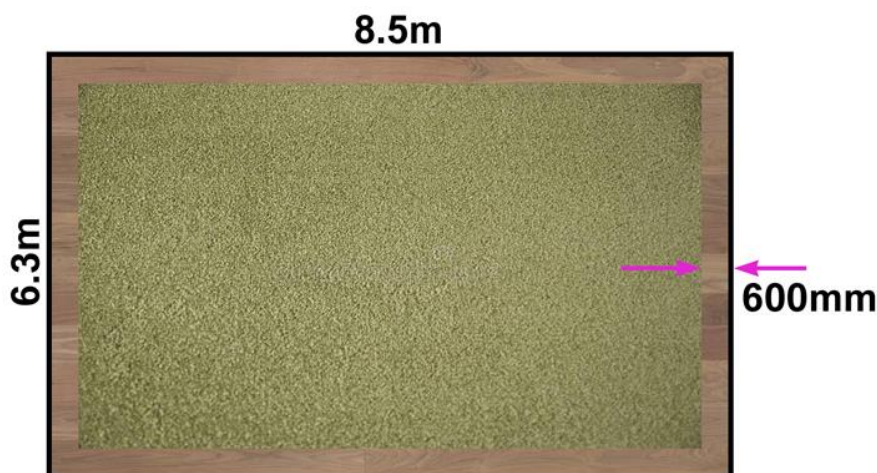
Calculation of Areas of Shapes

Sometimes an area calculation must be made where the object or shape is not one of the common shapes listed. Sometimes it is made up from a combination of shapes.

Example:

An office **8.5m by 6.3m** is to be fitted with a carpet, so as to leave a surround **600mm** wide around the carpet. What is the area of the surround?

With a problem like this, it is often helpful to sketch a diagram.



The area of the surround = office area - carpet area

$$\begin{aligned}
 &= (8.5 \times 6.3) - (8.5 - 2 \times 0.6) (6.3 - 2 \times 0.6) \\
 &= 53.55 - (7.3) (5.1) \\
 &= 53.55 - 37.23 = 16.32\text{m}^2
 \end{aligned}$$

Note that 600mm had to be converted to 0.6m.

Don't forget to include units in the answer e.g. m².

We may need to find the area of an object that is a combination of shapes:

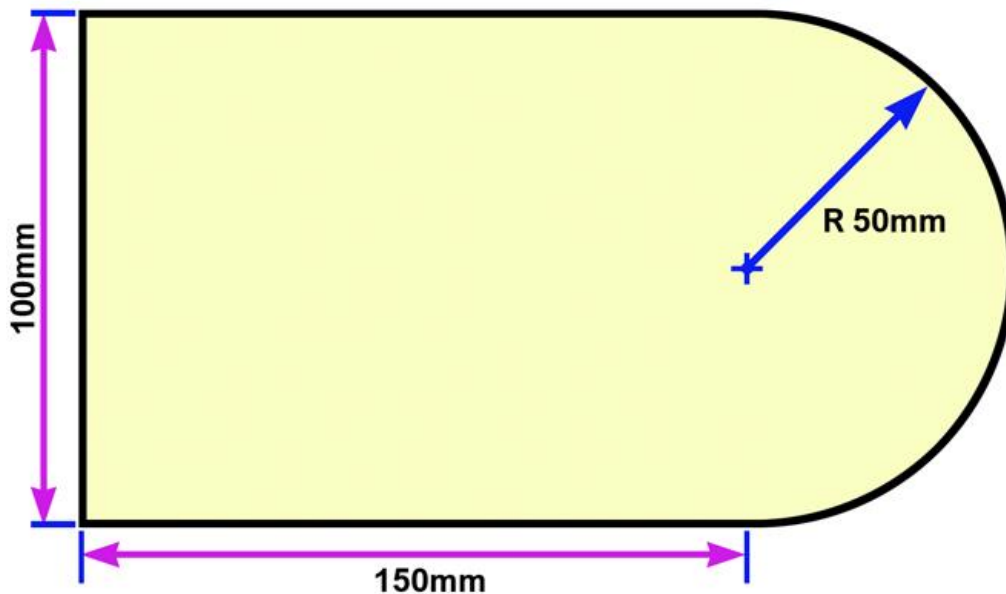


Fig 04. 27 -Combination Shapes

In this case the shape comprises a rectangle and a semi-circle.

- **The rectangle has dimensions 150mm x 100mm.**
- **The semi-circle has a diameter of 100mm.**

Total area is the sum of the two individual areas.

$$\text{Area} = (100 \times 150) + (\pi 50^2) / 2$$

$$\text{Area} = 15000 + 3,925 = 18925 \text{mm}^2$$

VOLUMES

Solids are objects that have three dimensions: **length**, **width** and **height**. Having the ability to calculate volume enables you to determine the capacity of a fuel tank or reservoir, calculate the capacity of a cargo area or work out the volume of a cylinder.

Volumes are calculated in cubic units such as **cubic centimetres**, **cubic metres**, **cubic inches** etc. However, volumes are easily converted to other terms, such as **litres**. For example:

A cubic metre contains 1000 litres of liquid.

Instead of squares, we now consider cubes. This is a 3-dimensional concept and the typical units of volume are **cubic metres (m^3)**.

If we have a box

- **Length 4m**
- **Width 3m**
- **Height 2m**

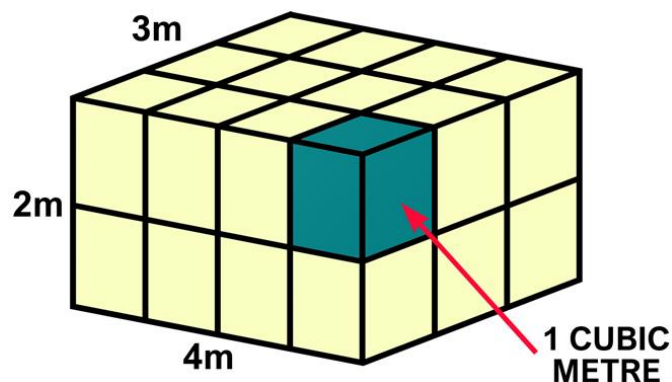


Fig 04. 28 - Cubes

We see that the total volume = 24 cubic metres ($24m^3$).

Each layer contains

$$4 \times 3 = 12 \text{ cubes.}$$

There are 2 layers. Hence the volume is:

$$12 \times 2 = 24m^3$$

Basically, therefore, when calculating volume, it is necessary to look for three dimensions, at **90°** to each other, and then multiply them together. For a box - type shape, multiplying **length x width x height = volume**.

For irregular or particular shapes, different techniques or approximations can be used, or sometimes a specific formula may exist.

Example:

$$\text{Volume of Cylinder} = \pi R^2 h \quad (R = \text{radius, } L = \text{height})$$

$$\text{Volume of a Cone} = \frac{1}{3} \pi R^2 h$$

$$\text{Volume of a Sphere} = \frac{4}{3} \pi R^3$$

Note that all these formulae contain **3** dimensions so that when multiplied, a volume will result.

$$\text{e.g. } R^2 h = R \times R \times h \quad \text{or} \quad R^3 = R \times R \times R$$

If you have not got **3** dimensions, you have not got a volume!

Example:

What is the cubic capacity of a **2** cylinder engine, with a bore of **77mm** and a stroke of **89mm**?



BORE = DIAMETER = 77mm

STROKE = HEIGHT = 89mm

VOLUME OF CYLINDER = AREA OF CIRCLE x HEIGHT

Fig 04. 29 - Volume of a Cylinder

$$\text{Volume of 1 cylinder} = \pi \left(\frac{77}{2}\right)^2 \times (89)$$

$$\text{Volume of 1 cylinder} = 414440 \text{ mm}^3$$

$$\text{Volume of 2 cylinders} = 828880 \text{ mm}^3$$

Note

In this example, the dimensions have been given in mm.

The volume would normally be given in cm³.

To convert mm³ to cm³, divide by (10)³

$$\therefore 828880 \text{ mm}^3 \text{ becomes } 828.88 \text{ cm}^3$$

When calculating areas or volumes, remember the basic formulas, but be ready to spot when an area or solid body is a combination of basic shapes that can be added or subtracted.

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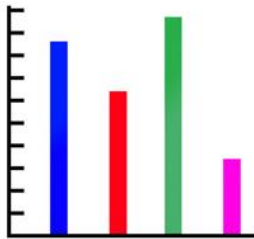
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IKAROS

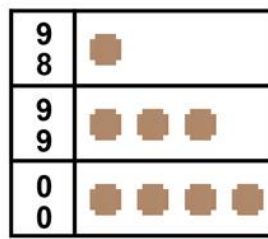


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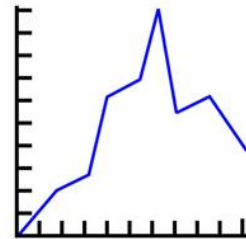
BAR GRAPH



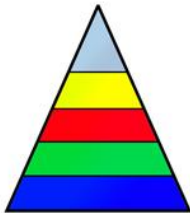
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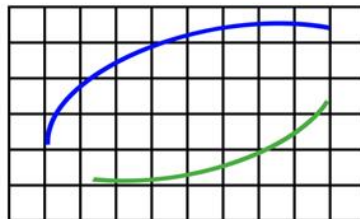
BROKEN LINE GRAPH



TRIANGLE CHART



CONTINUOUS CURVE LINE GRAPH



PIE CHART



PART 66 MODULE 6 B1/B2

SECTION 05

GRAPHS

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GRAPHS

Introduction

Graphs are a pictorial method of displaying numerical data that enables you to quickly visualise certain relationships, complete complex calculations and predict trends. The data can be presented in many different ways as shown below, and most data can be presented in any format. However, care should be taken when selecting a format to use, some formats are better suited to particular types of data or data sets.

For example, if have a whole amount divided into known proportions, then this is better presented as a pie chart; if we have a list of scores in a test, then a bar graph is better. If we are plotting temperature with respect to time then a continuous line graph is better,

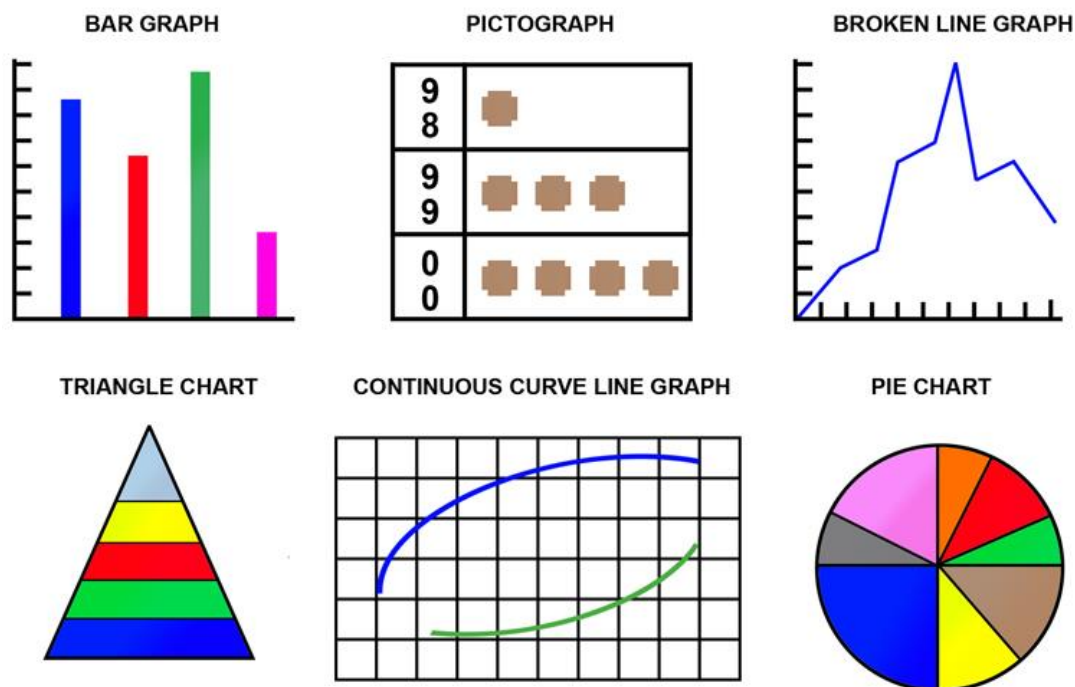


Fig 05. 1 - Types of Chart

Construction

In order to construct graphs effectively, some simple rules should be followed. First of all, present the data in a clear, tabular form. The data will generally comprise 2 variables, one that is being varied, the independent variable, and the one that changes as a result of the variation, the dependent variable (its value depends on the value of the other).

For example, an experiment was conducted, where a volume of gas was heated. As the temperature of the gas increased, it was noted that the gas expanded: its volume increased. The first quantity, the temperature, is the independent variable and the second quantity, the volume, is the dependent variable.

The next stage is to plan the use of the graph-paper so as to present the graph in the clearest manner possible.

The graph constructed by plotting a series of points, each one representing a particular value of the independent and corresponding dependent variable. So the graph must be drawn so that each value appears (or fits) on the paper.

Axis and Scales

Before “**plotting**” the points, the two axes must be drawn, and the scales chosen.

The **horizontal (x-axis)** will represent the independent variable and the **vertical (y-axis)** the dependent variable. The scales cross at the **origin O**.

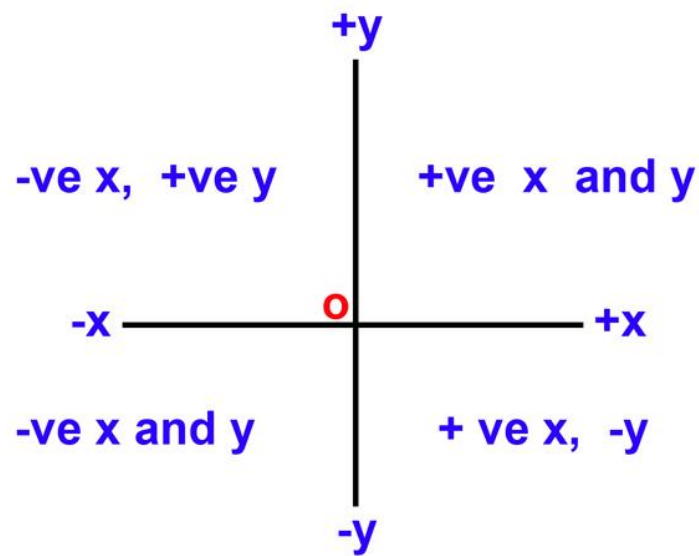


Fig 05. 2 - Axis and Scales

There is no merit in drawing small graphs. Choose scales so that completed graph fits the sheet of graph paper.

- Look at the largest right-hand, and the smallest left-hand values that will be plotted along the x-axis.
- Subtract the LH value from the RH value to give a range of values (= some number of units).
- Study the graph paper to find how many large squares there are from left to right.
- Now divide the value found by the subtraction, by the number of large squares.

This should give an idea of a suitable scale.

That is, so many units should be represented by **1 large square** along the **x-axis**.

The most useful scales are **1, 2, 5, 10, 20, 50 units** etc. etc to **1 large square**.

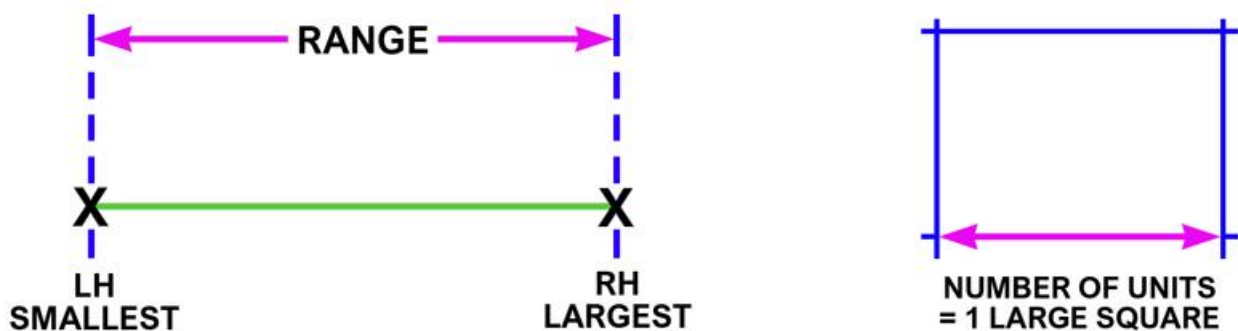


Fig 05. 3 - Plotting Graphs

The same procedure is used for the **y-axis**. Subtract the smallest (lower) value from the largest (upper value) to give the **Range**, divide by the number of large squares between top and bottom of the paper.

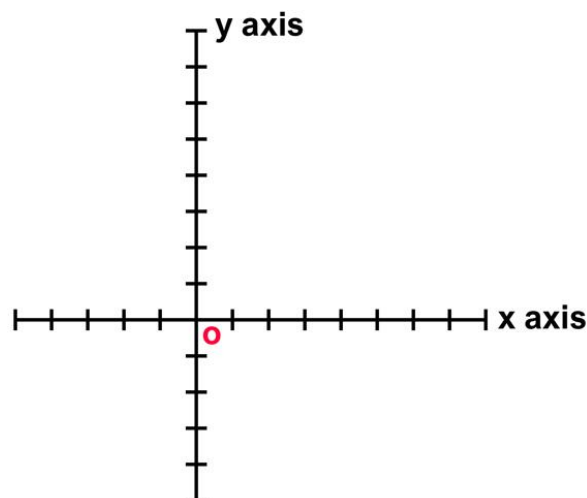


Fig 05. 4 - x and y Axis

Having done this, draw the **2 axes**, and mark off the units, using your chosen scales.

The graph paper has now been prepared for the object of the exercise, i.e. to transfer the data from the table to the graph.

The transfer is very simple, take one value of the independent variable and draw a (faint) line to coincide with its value along the **x-axis** so as to intersect with a similar line drawn from the **y-axis** for its corresponding dependent value.

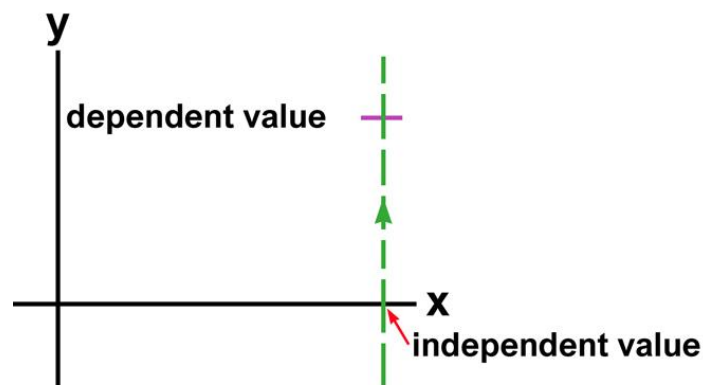


Fig 05. 5 - Setting Values

The intersection represents one plotted point of the graph.

The procedure is repeated for each pair of values in turn. When all the points have been plotted, a continuous line is drawn through the points.

The way in which the line is drawn depends on the nature of the data. It is probably true to say that most mathematical or scientific data change gradually or progressively - they may form a definite relationship. In this case, do not join the points with a series of straight lines.

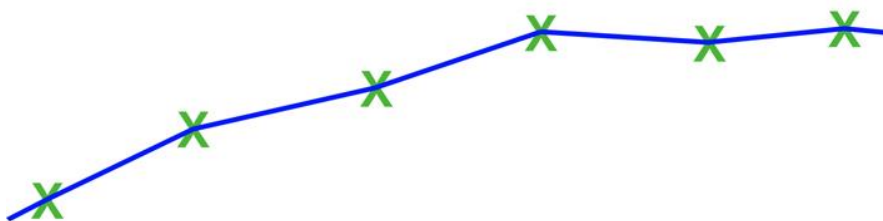


Fig 05. 6 - Straight Lines

But try to draw a continuous smooth line.

This probably means that the line only goes through some (not all) of the points - don't worry; experimental or plotting errors can occur.

There should be roughly the same number of points on both sides of the smooth curve. Sometimes, it is fairly obvious that a straight line is the (most) reasonable 'fit' to the point, and this is often the case for simple scientific experiments.

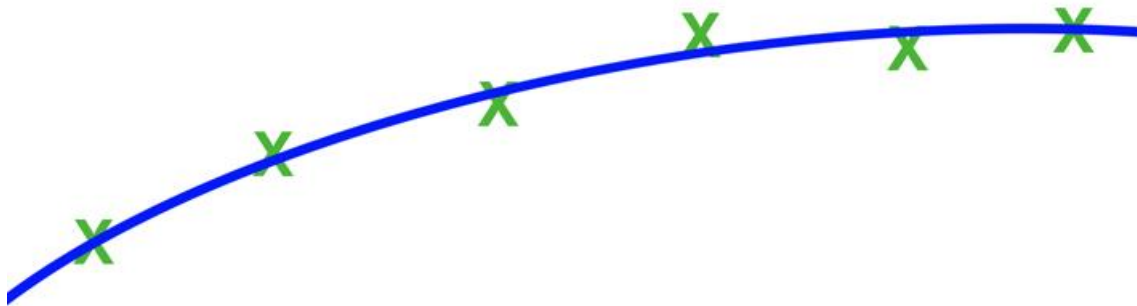


Fig 05. 7 - Curved Lines

GRAPHS AND MATHEMATICAL FORMULAE

This course is designed for engineers, not mathematicians and so maths is viewed as a servant, not a master. Later, it will be seen that one physical quantity will vary as another quantity varies, with the two linked by some mathematical law or equation.

An example is that the **drag force (D)** varies according to the square of the **airspeed (V)**. Expressed as a formula:

$$D = kV^2$$

This relationship can be plotted in graphical form, and it is reasonable to presume that it would be of the same form as the maths relationship of $y = x^2$ where y is considered as a function of x $y = f(x)$

There are many mathematical functions, examples might be:

$$y = mx, \quad y = x^2, \quad y = x^3, \quad y = \sin x$$
$$y = e^x, \quad y = \cos x \quad \text{etc. etc}$$

This topic looks at the shape and characteristics of these functions when expressed graphically, so that a simple link can be made with physical phenomena, which demonstrates similar characteristics.

When a mathematical function is plotted, certain shapes evolve characteristic of that function. If, following an experiment during which data is gathered, that data creates similar shapes, then a presumption linking formula and experiment may be made.

Function and Shape

The variable **y** is often described as a function of **x**. Here several different functions are considered graphically.

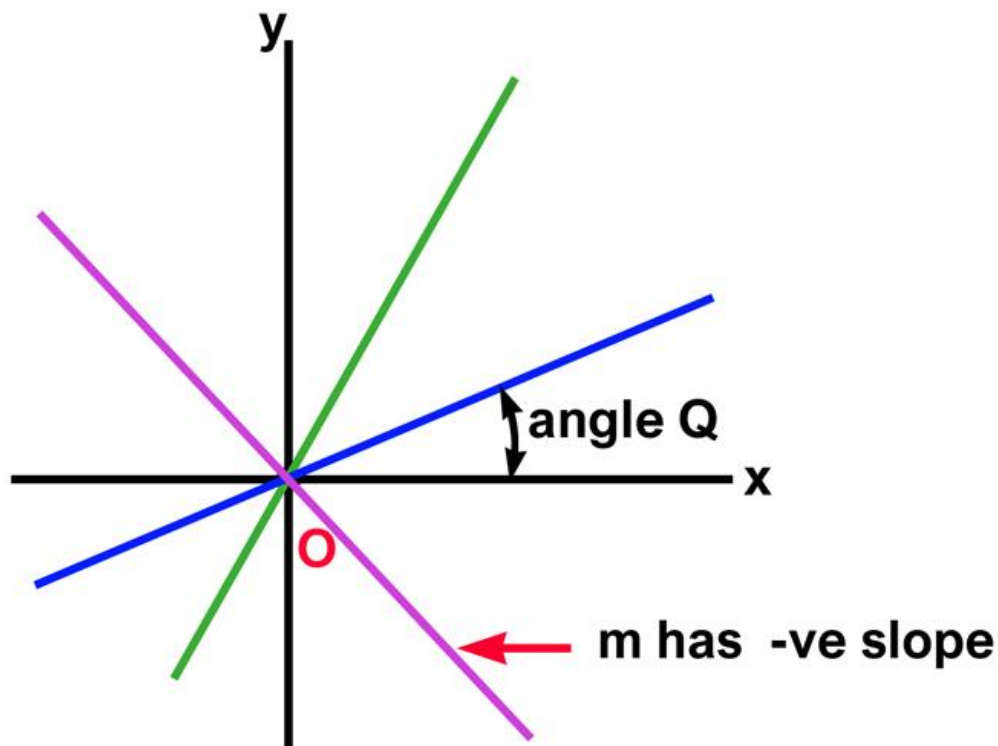


Fig 05. 8 - Functions

Function $y = mx$ where **m** is some constant coefficient.

$y = mx$ gives a straight line, passing through the **origin O**.

m is the slope of the graph (and $= \tan Q$ the greater the value of **m**, the steeper the slope).

Obviously for a straight line, the slope is constant for a constant value of **m**.

If **m** is **-ve**, the line slopes as shown. (if $m = 0$, the 'line' $Y = 0$ coincides with the **x-axis**).

Function

$$y = mx + c$$

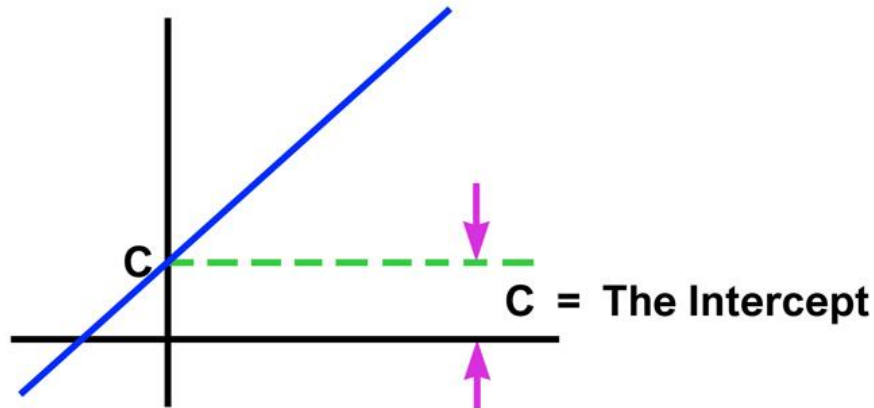


Fig 05. 9 - The Intercept

This is a variation of $y = mx$.

C is a constant, and is clearly the value of **y** when **x = 0**. ($y = m \cdot 0 + c = c$). This value of **c** measured along the **y axis** is known as the '**Intercept**'.

Function

$$y = kx^2 \text{ where } k \text{ is some constant.}$$

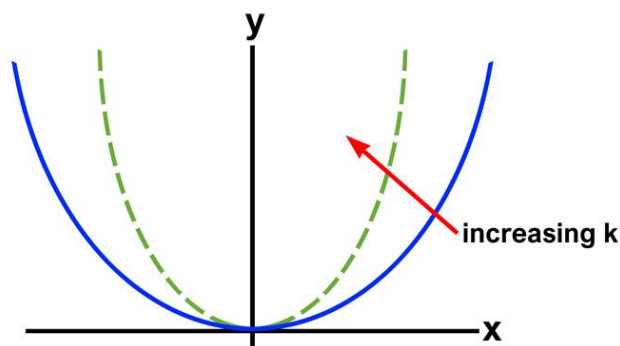


Fig 05. 10 - Parabola

This gives a curve, known as a parabola. As '**k**' increases the value of kx^2 also increases. Note that the slope is no longer constant. This is a function which is commonly found in physical situations.

Function

$y = kx^3$ etc.

This is the characteristic shape. Note that the graph has Turning points, where the slope changes from **+ve** to **-ve** and vice versa.

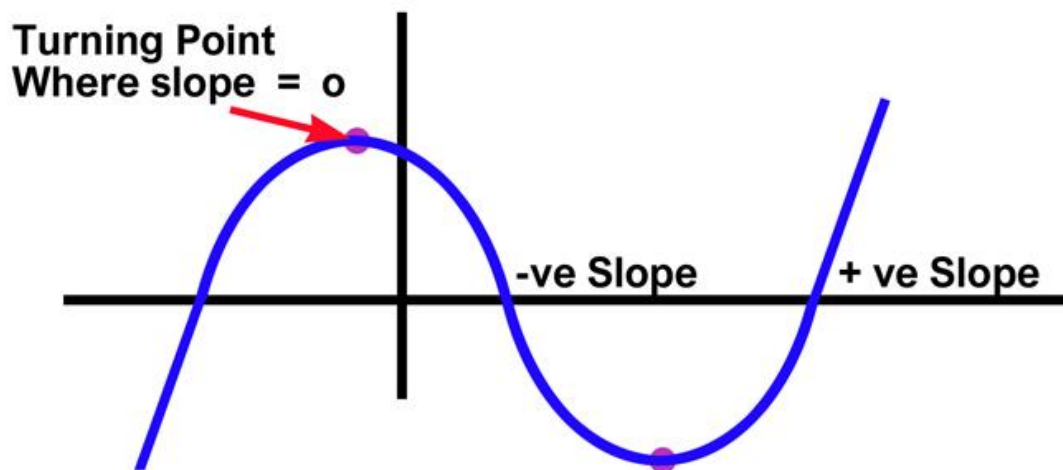


Fig 05. 11 - Turning Points

Functions within this family are less likely to be encountered during this course.

Function

$$y = \sin x \text{ and } y = \cos x$$

Both of these functions are repetitive but the word used to describe such behaviour is periodic (in this case, the period is **360° or 2π radians**).

Note that the cosine graph 'leads' the sine graph by 90° ($\frac{\pi}{2}$ radians) when such behaviour occurs, it is often referred to a '**phase difference**'. These graphs are often found, particularly in electrical work.

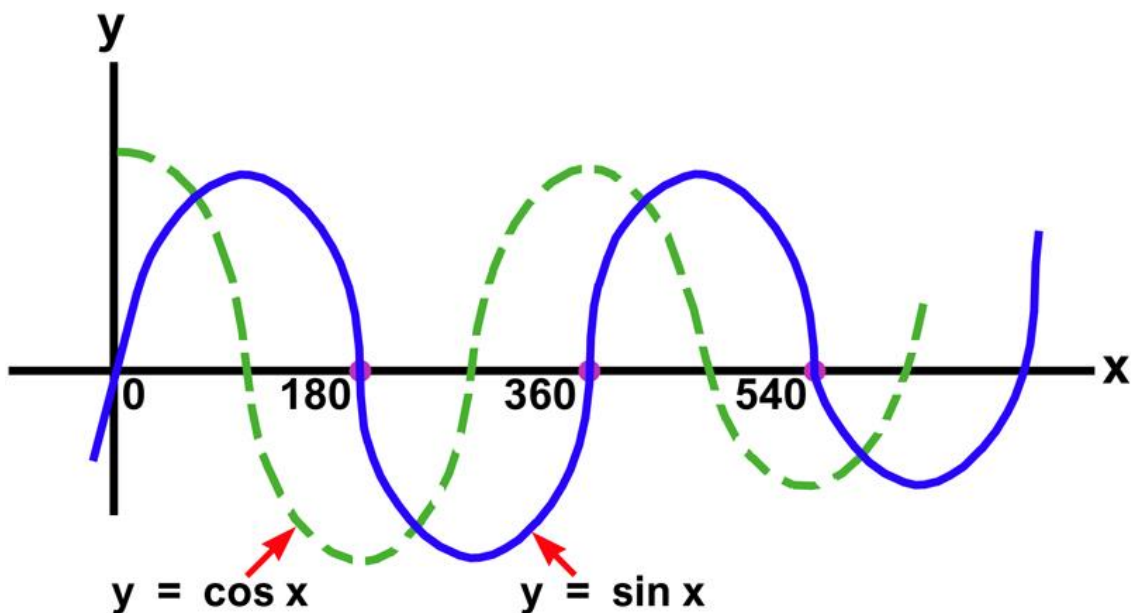


Fig 05. 12 - Function $y = \sin x$ and $y = \cos x$.

Function

$$y = e^x, \quad y = e^{-x}, \quad y = 1 - e^{-x}$$

$y = e^x$ is known as the **Exponential function**. It is also often found in engineering applications. Some variations on the basic function are also shown.

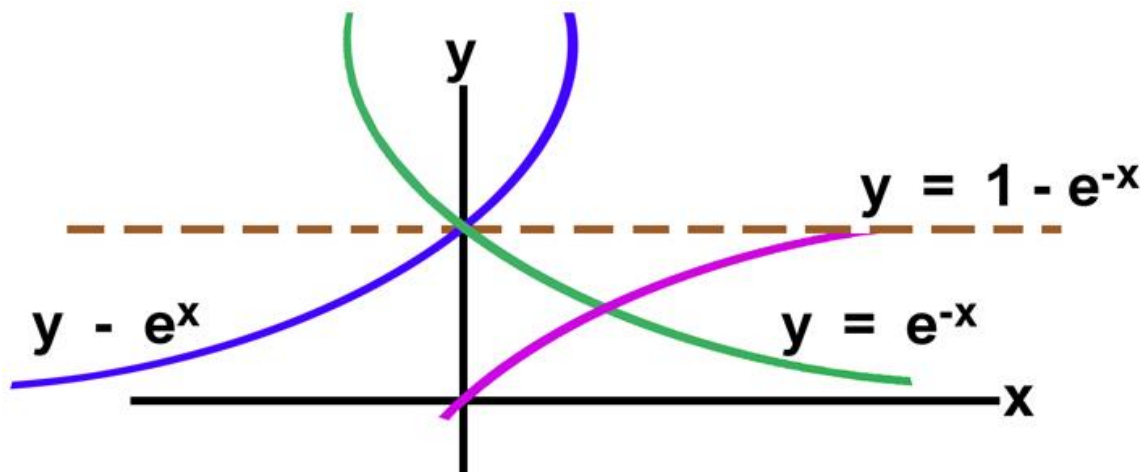


Fig 05. 13 - Function $y = e^x$

Reference has already been made to the slope of a graph. Straight lines have a constant slope. Curves have variable slopes, and often include turning points (often termed **maxima** and **minima**).

Mathematicians determine slopes by using a branch of mathematics called '**Calculus**' – a later topic. Engineers are often interested in slope, because depending on the variables, the slope itself represents a physical quantity – more about this in the Physics module.

Area under a Graph

The area under a graph is also often useful and may represent a physical quantity.

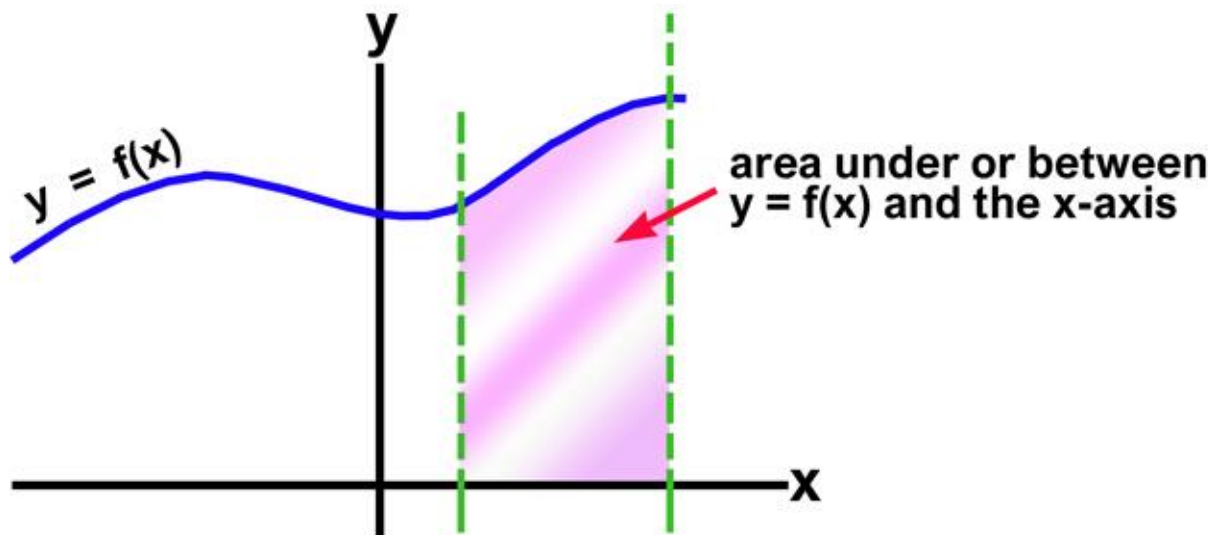


Fig 05. 14 - Area Under a Graph

The area can be calculated by:

- **Considering simple shapes and approximating**
- **Counting squares**
- **Using calculus**

NOMOGRAPHS

The need to show how two or more variables affect a value is common in the maintenance of aircraft. Nomographs are a special type of graph that enable you to solve complex problems involving more than one variable.

Most nomographs contain a great deal of information and require the use of scales on three sides of the chart, as well as diagonal lines.

In fact, some charts contain so much information, that it can be very important for you to carefully read the instructions before using the chart and to show care when reading information from the chart itself.

Illustrated is a fairly typical graph of **three** variables:

- **Distance**
- **Speed**
- **Time**

If any two of the three variables is known, the approximate value of the third can be quickly determined. In this example, the dotted line indicates a known speed and time. The resulting distance travelled can be extracted from the graph at the point where these two dashed lines meet.

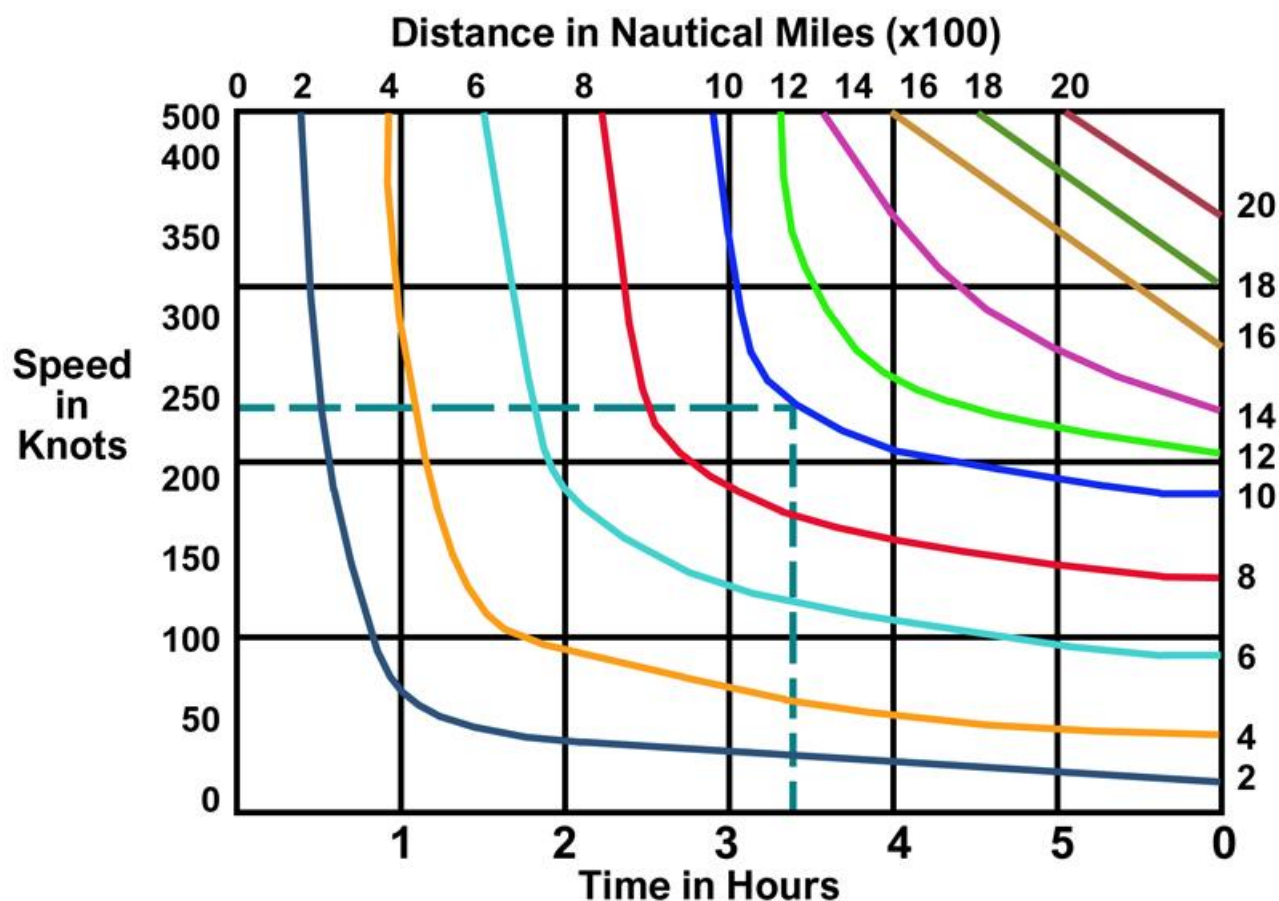


Fig 05. 15 - Nomograph

Whilst this nomograph is much too small for accurate computation, it can be seen that when travelling at around 250 knots for three and a half hours, you would travel just under 1000 nautical miles.

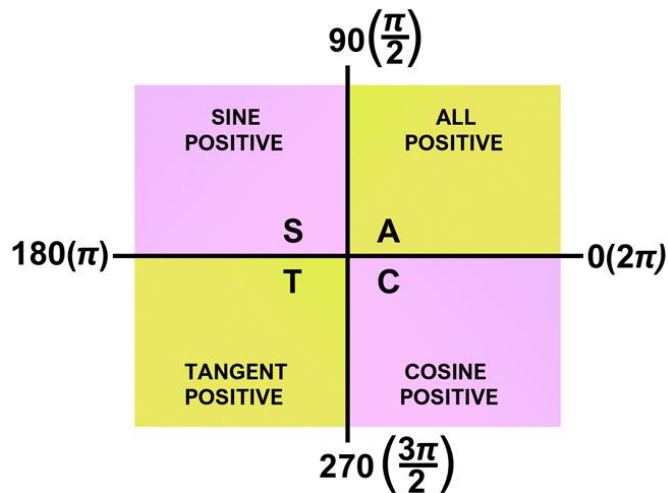
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PART 66 MODULE 6 B1/B2

SECTION 06

TRIGANOMETRY

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TRIGONOMETRY

Introduction

Basic trigonometry involves expressing the angles of a right-angled triangle in relation to lengths of the sides of the triangle.

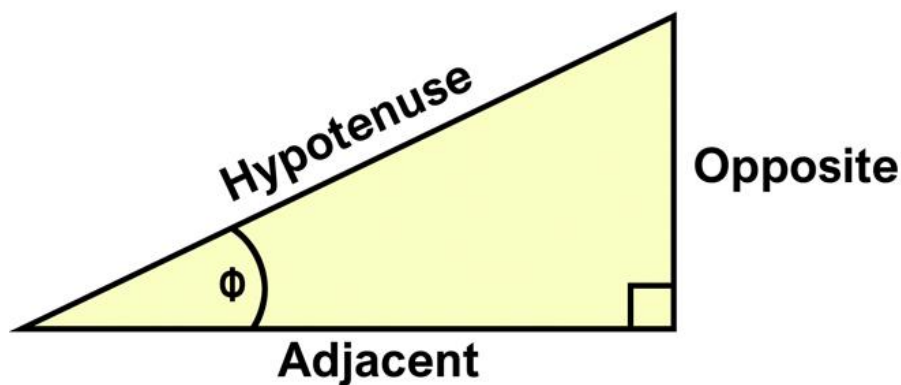


Fig 06. 1 - Right-Angled Triangle

The ratio of the opposite side length to the hypotenuse length in the diagram is termed the "**sine**" of the angle ϕ .

$$\sin \phi = \frac{\text{Opposite}}{\text{Hypotenuse}} \quad \frac{o}{h}$$

$$\cos \phi = \frac{\text{Adjacent}}{\text{Hypotenuse}} \quad \frac{a}{h}$$

$$\tan \phi = \frac{\text{Opposite}}{\text{Adjacent}} \quad \frac{o}{a}$$

These ratio's must be remembered!

(Some students find the mnemonic "**Sohcahtoa**" to be helpful in this respect).

These ratios are used very extensively in Maths and Science and very many modifications to the basic ratio have been evolved.

How can these ratios be used in practice?

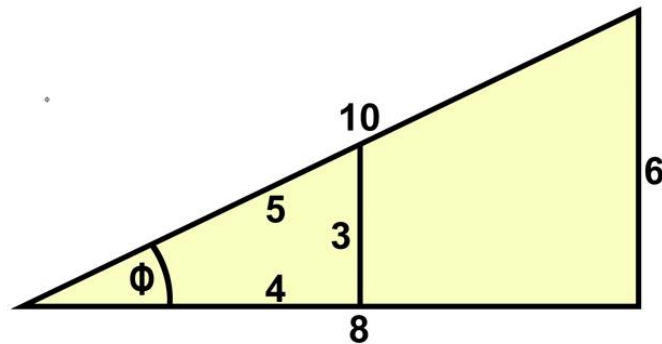


Fig 06. 2 - Triangle

Consider a triangle with side lengths **3, 4, 5 OR 6,8,10** as shown.

From our definition of sine, $\frac{3}{5} = 0.6 = \text{sine } \phi$
 $\frac{4}{5} = 0.8 = \text{cosine } \phi$

Now while it is obvious that ϕ is proportional to the side lengths, what is its actual value in degrees?

e.g. if **0.6** is input into a calculator and the **$\sin^{-1}$** button is operated, the screen display will be **36.86989765°**.

The actual calculation of sine, cosine and tangent is beyond the scope of this course, but the values of each ratio and the corresponding angle have been compiled in tabular form, but can be found using a scientific calculator.

If **0.8** is input and the **$\cos^{-1}$** button operated or if $\frac{3}{4} = 0.75$ and the **$\tan^{-1}$** button operated the same **36.86989765** will be displayed. Conversely, if **36.86989765** is input, and the **$\sin$** button is operated, **0.6** will be displayed.

TRIGONOMETRICAL CALCULATIONS & FORMULA

Introduction

Earlier we considered the basic trigonometry functions. They can now be applied to practical situations.

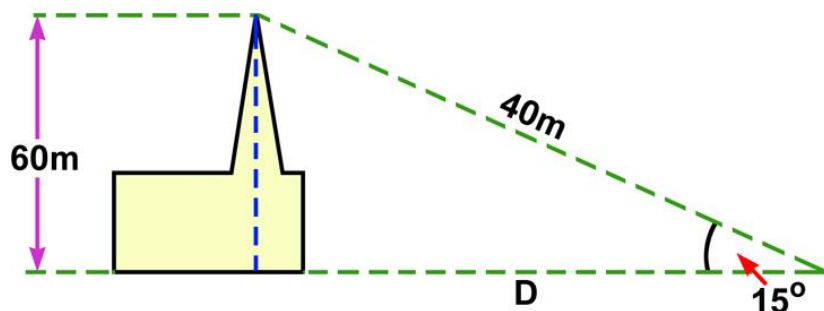


Fig 06. 3 - Church Spine

Example

A church spine is known to be **60** metres high. When the top is viewed through a theodolite, the angle between the line-of-sight and the horizontal is **15°**. How far is the theodolite from the base of the spine?

The **distance D** is the unknown quantity. **Angle 15°** and **side (height) 60m** are known.

Therefore, an equation can be formed:

$$\frac{60}{D} = \frac{O}{A} = \tan 15^\circ$$

$$\text{Transposing } D = \frac{60}{\tan 15^\circ}$$

Using the calculator, **60 ÷ tan 15 = 223.9 metres**.

This illustrates the basic principle when solving trigonometry problems.

Sketch a diagram if necessary, identify the known and unknown values, and then express them in terms of the sides of the triangle and the corresponding angle.

The basic trigonometry ratios were explained with reference to a right-angled triangle. But their use can be extended for use with any triangle.

Example

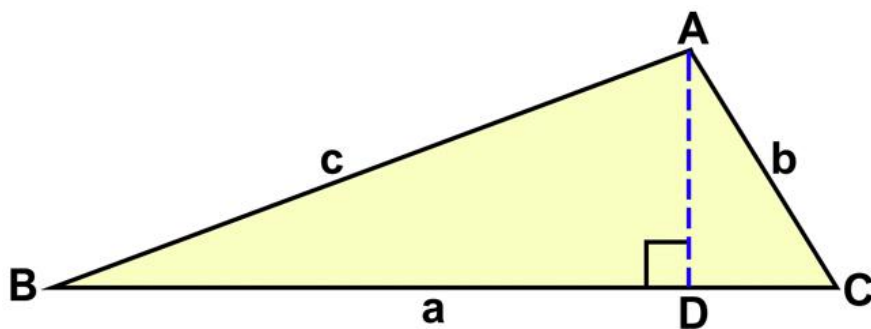


Fig 06. 4 - Example 1

ABC is any triangle.

Suppose a line **AD** is drawn so that **angle BDA = angle CDA = 90°**. **AD** is now the height of the triangle.

$$\text{The area of the triangle} = \frac{1}{2} a \times AD(c)$$

$$\text{but } \frac{AD}{b} = \sin C \left(\frac{\text{opposite}}{\text{hyotenuse}} \right)$$

$$\text{Therefore } AD = b \sin C \quad \textcircled{2}$$

Substituting $\textcircled{2}$ in $\textcircled{1}$

$$\text{The area of the triangle} = \frac{1}{2} a.b \sin C$$

Using a similar method it can be shown that the area of a triangle is also:

$$\frac{1}{2} b.c \sin A = \frac{1}{2} a.c \sin B$$

Using these last two equations we can derive the sine formula.

$$\frac{1}{2} \cdot b \cdot c \cdot \sin A = \frac{1}{2} \cdot a \cdot c \cdot \sin B$$

$$b \cdot c \cdot \sin A = a \cdot c \cdot \sin B$$

$$b \cdot \sin A = a \cdot \sin B \quad (\text{dividing through by } c)$$

$$\therefore \frac{b}{\sin B} = \frac{a}{\sin A}$$

Another useful formula is the Cosine formula. Again it applies to any triangle **ABC** and has three forms.

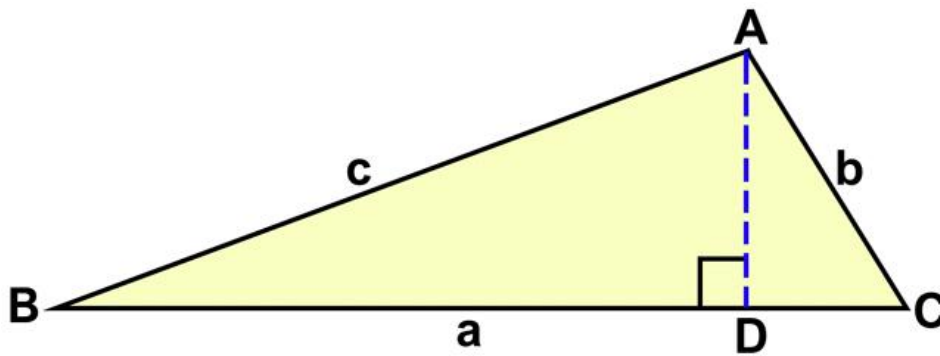


Fig 06. 5 - Example 2

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

$$\cos B = \frac{a^2 + c^2 - b^2}{2ac}$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

(These formula can easily be proved by drawing AD perpendicular to BC, and using Pythagoras).

CONSTRUCTION OF TRIGONOMETRICAL CURVES

If **radius OP** is rotated anticlockwise, the angle **θ (POA)** increases and the value of **sine $\div$** also increases (because **AP** increases in relation to **OP**).

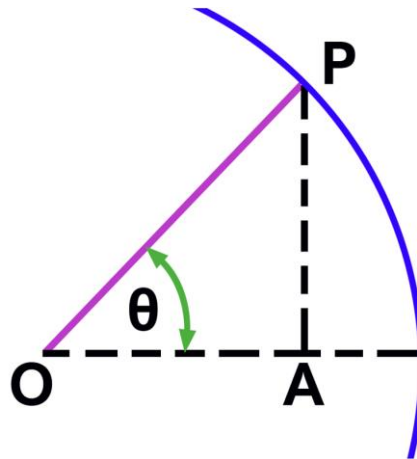


Fig 06. 6 - Construction of Curves

If the radius **OP** has a length of 1 unit,

$$\text{sine } \theta = \frac{AP}{1} = AP \text{ (the length AP)}$$

If a graph of **sine $\div$** (length **AP**) is plotted against **angle θ** , the typical curve results.

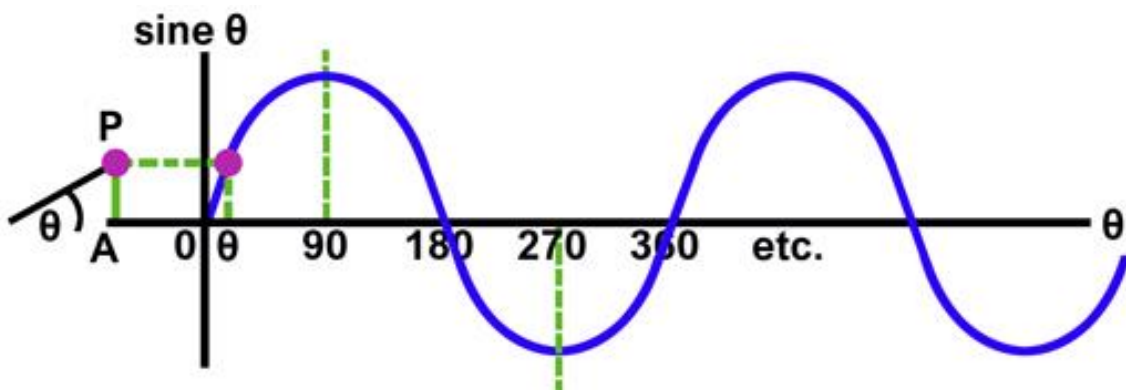


Fig 06. 7 - Graph of Sine

Note the repetition every revolution (**360°**) and that the values of **sine $\div$** range between **+1 and -1**.

The graph for **cosine θ** is similar but displaced by **90°** .

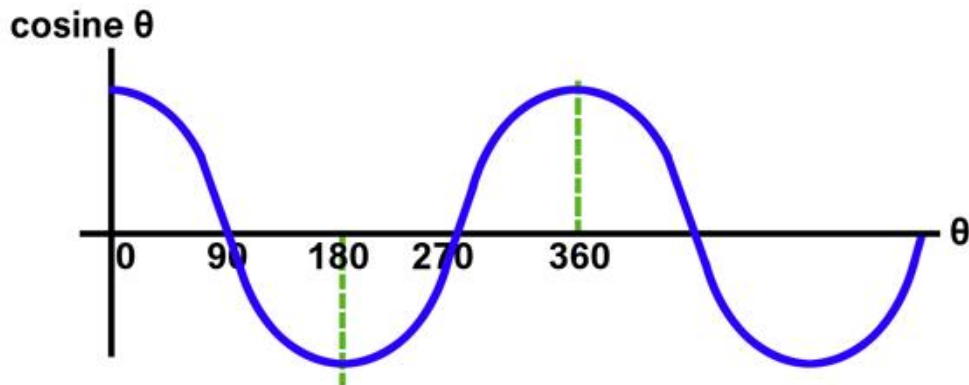


Fig 06. 8 - Graph for Cosine θ

The graph for **tangent θ** is deduced from the other two curves

$$\tan\theta = \sin\theta/\cos\theta$$

At **90°** and **270°** , the value of **$\tan \theta$** becomes infinity.

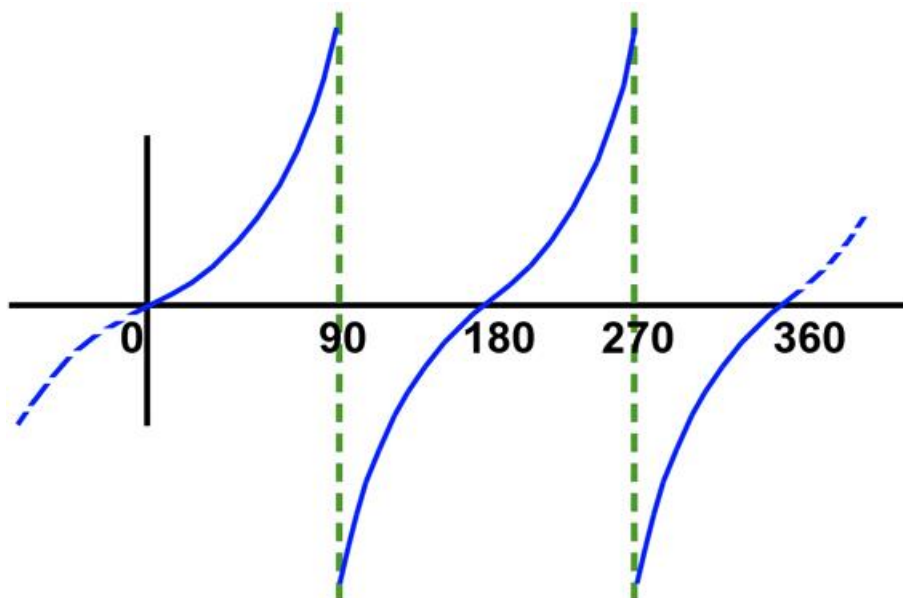


Fig 06. 9 - Graph for Tangent θ

Values in 4 Quadrants

Inspection of the **sine** and **cosine curves** show that the values change from **+ve to -ve to +ve etc.**, as **angle θ increases**. It is important to have an idea how these changes are linked to the approximate value of θ .

This diagram shows how the values of **sine**, **cosine** and **tangent** take **+ve or -ve** values, depending with the value of θ , within one of the four quadrants.

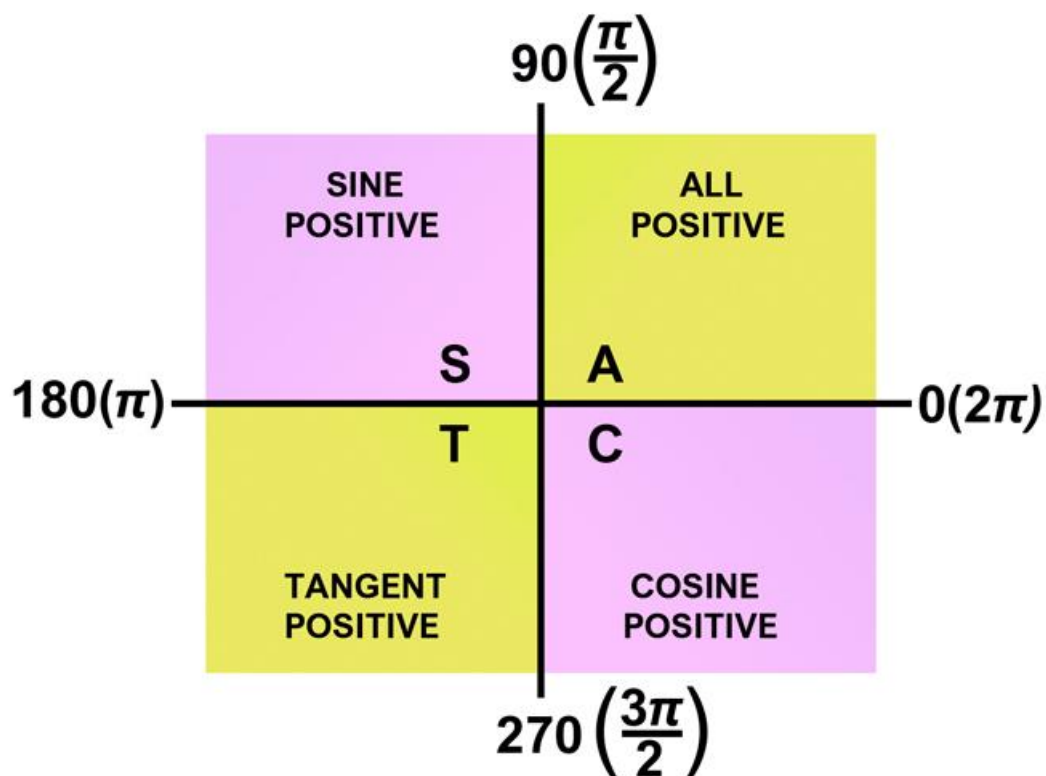


Fig 06. 10 -Values in 4 Quadrants

CO-ORDINATE GEOMETRY

Introduction

Geometry has previously considered certain well-known regular shapes, e.g. circles, rectangles and triangles, and studied their properties. These studies have considered the shape in isolation, i.e. without reference to any particular datum.

Co-ordinate geometry extends these studies by introducing '**Datums**', and then expressing the position of the significant features of shapes with reference to their datum. The datums we chose are usually the **x, y, z axes** we use in graphs.

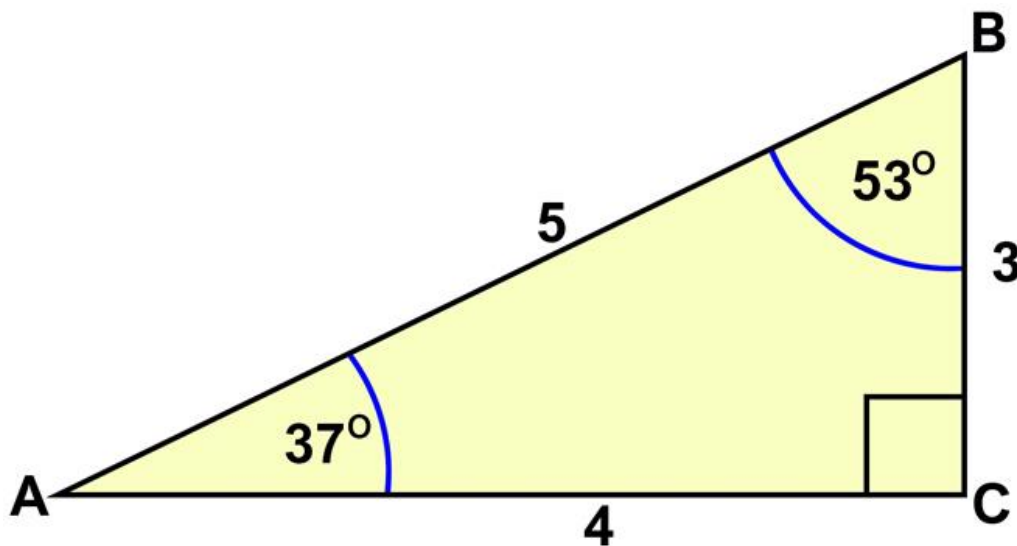


Fig 06. 11 - Datums 1

Example

Suppose we had a right-angled triangle, sides and lengths **3**, **4** and **5** units. We know that the angles are approximately **37°**, **53°** and **90°**.

We might chose to place the triangle in our **x, y plane**, where point **A** is **2** units along the **x axis**, and **1** unit along the **y axis**. The co-ordinates of point **A** then become **(2.1)**.

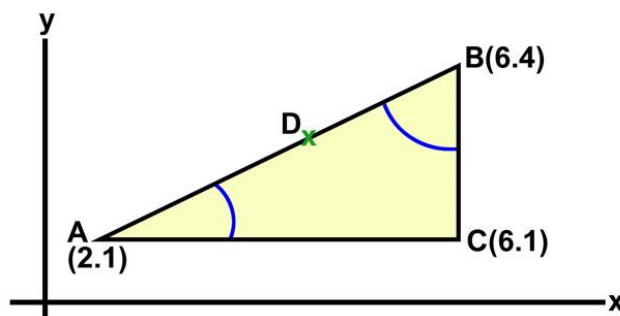


Fig 06. 12 -Example 1

As long as **AC** is drawn parallel to the **x-axis**, **point C** becomes **(6.1)** and **point B** becomes **(6.4)**. If we introduce **point D** as the mid-point along **AB**, it is clear that the co-ordinates of **D** are **(4, 2.5)**. If we were to fix point **A**, but rotate the triangle, the co-ordinates of **B** and **C** would change, even though the length of the side remains unchanged.

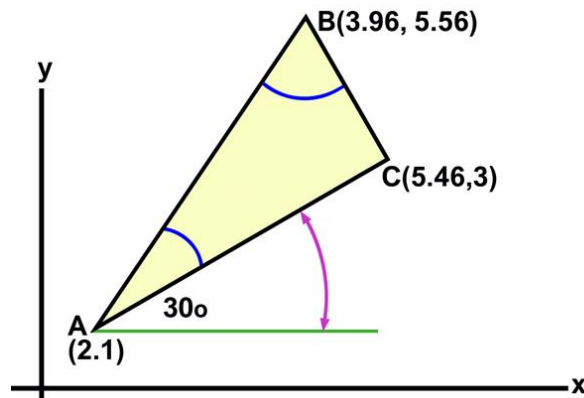


Fig 06. 13 -Example 2

Point C moves from **(6.1)** to become **(5.46, 3)** and **point B** moves from **(6.4)** to become **(3.96, 5.56)**.

Note

The student will not be required to calculate the change in co-ordinates but to appreciate how a change of position is accompanied by a change in co-ordinates, even though the basic shape is unchanged.

In this example, the point **A, B and C** have co-ordinates which are positive integer values, but they could have been given symbols, such as

(x_a, y_a) (x_b, y_b) and (x_c, y_c)

In further Maths studies, this would be more usual.

POLAR / RECTANGULAR CO-ORDINATES

Introduction

In the previous chapter, the concept expressing geometrical shapes by using (x, y) co-ordinates was considered.

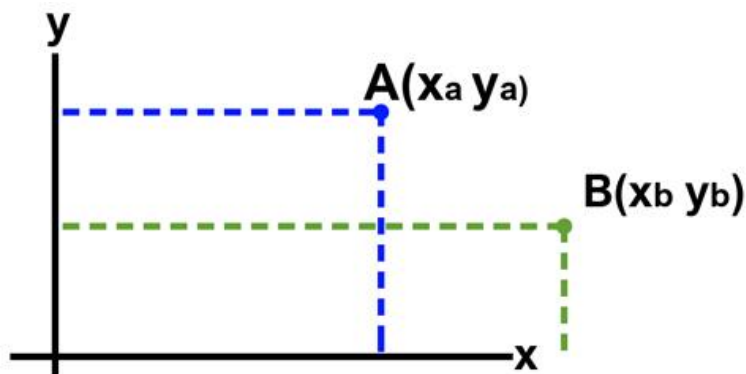


Fig 06. 14 -Rectangular Co-ordinates

Because their position is established with reference to the two axes **x and y**, where the intersection of the two reference values completes a rectangle, such co-ordinate are known as **Rectangular Co-ordinates** (they are sometimes known as **Cartesian Co-ordinates**).

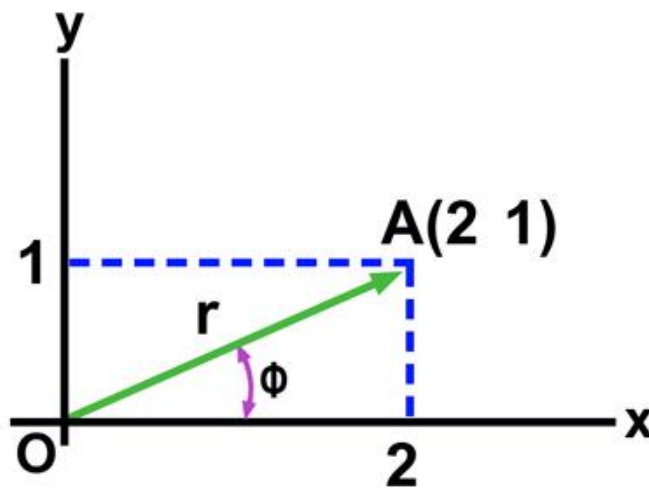


Fig 06. 15 -Cartesian Co-ordinates

Another co-ordinate system exists, which uses a different system of datum's. In this system, the datum's are the origin, **point O** and the **x axis**, and the **position A**, which was expressed as **(2, 1)** using **(x y)** co-ordinate will now be expressed in terms of the distance from **O (the 'r' co-ordinate)** and the angular displacement of the line **OA** with reference to the **x axis** (the **co-ordinate**).

This method of expressing position(s) in terms of **r and ϕ** is known as the **Polar co-ordinate system. (r, theta)**

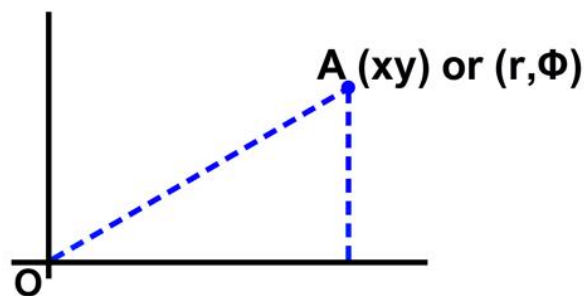


Fig 06. 16 -Polar Co-ordinate System

Although the student will not be required to perform extensive calculations using either system in this module, a basic appreciation is necessary, which should include the ability to relate one system to the other.

For Example

To convert Rectangular to Polar:

$$\begin{aligned} r^2 &= x^2 + y^2 & r &= \sqrt{x^2 + y^2} \\ \tan \phi &= y/x & \phi &= \tan^{-1} (y/x) \end{aligned}$$

To convert Polar to Rectangular:

$$\begin{aligned} x &= r \cos \phi \\ y &= r \sin \phi \end{aligned}$$

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